

Algebra III

Lecture notes

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Tablet reading edition

The material draws on Serge Lang's *Algebra*, Jean-Pierre Serre's *Linear Representations of Finite Groups*, and Charles A. Weibel's *An Introduction to Homological Algebra*.

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1 Semisimple modules and rings

1.1 Linear maps and matrices

Definition. For a ring A , $\text{Mat}_{m \times n}(A)$ consists of arrays (a_{ij}) . Multiplication is $(AB)_{ik} = \sum_j a_{ij}b_{jk}$. The opposite ring A^{op} has multiplication $a \cdot_{\text{op}} b = ba$.

Proposition. For left modules, images of standard basis vectors give additive isomorphisms

$$\text{Hom}_A(A^n, A^m) \simeq \text{Mat}_{m \times n}(A^{\text{op}}).$$

Using opposite multiplication makes this correspondence compatible with composition.

$$\begin{array}{ccc} \text{Hom}_A(A^m, A^\ell) \times \text{Hom}_A(A^n, A^m) & \xrightarrow{\circ} & \text{Hom}_A(A^n, A^\ell) \\ \sim \downarrow & & \downarrow \sim \\ \text{Mat}_{\ell \times m}(A^{\text{op}}) \times \text{Mat}_{m \times n}(A^{\text{op}}) & \xrightarrow{\cdot} & \text{Mat}_{\ell \times n}(A^{\text{op}}) \end{array}$$

Hence matrix multiplication is associative and distributive, $\text{Mat}_n(A)$ is a ring, and $a \mapsto aI_n$ is a ring map. A module M is a left module over $\text{End}_A(M)$ by evaluation; M^n is a $\text{Mat}_n(A)$ -module by matrix multiplication.

Proposition. For finite direct sums, the components of a map are $f_{ij} = \pi_i f \iota_j$, giving

$$\text{Hom}_A\left(\bigoplus_j F_j, \bigoplus_i E_i\right) \simeq \bigoplus_{i,j} \text{Hom}_A(F_j, E_i), \quad f = \sum_{i,j} \iota_i f_{ij} \pi_j.$$

In particular $\text{End}_A(E^n) \simeq \text{Mat}_n(\text{End}_A(E))$, compatibly with actions on E^n .

$$\begin{array}{ccc} \text{End}_A(E^n) \times E^n & \xrightarrow{\text{ev}} & E^n \\ \sim \downarrow & & \parallel \\ \text{Mat}_n(\text{End}_A(E)) \times E^n & \xrightarrow{\text{matrix action}} & E^n \end{array}$$

For $E = A$, right multiplication identifies $\text{End}_A(A) \simeq A^{\text{op}}$, recovering the first formula.

1.2 Simple modules and Schur's lemma

Definition. A division ring is a nonzero ring in which every nonzero element is invertible. Modules over a division ring K have bases, and all bases have the same cardinality, by the same basis-extension and exchange arguments as over fields. These are left K -vector spaces, with dimension denoted $\dim_K$.

Definition. A module E is simple if it is nonzero and has no proper nonzero submodules.

Theorem (Schur's lemma). *A nonzero map between simple modules is an isomorphism, and $\text{End}_A(E)$ is a division ring.*

Proof. The kernel and image of a nonzero map must be 0 and the whole target. $\square$

Proposition. For simple E , an isomorphism $E^m \simeq E^n$ implies $m = n$, by applying $\text{Hom}_A(E, -)$ and comparing dimensions over $\text{End}_A(E)$ (with the appropriate right-module structure). More generally, for pairwise nonisomorphic simple modules,

$$\bigoplus_i E_i^{n_i} \simeq \bigoplus_j F_j^{m_j}$$

forces the same simple isomorphism classes and multiplicities, up to reordering: Schur's lemma kills Hom between distinct classes, and the preceding dimension argument handles each class. Also

$$\text{End}_A\left(\bigoplus_i E_i^{n_i}\right) \simeq \prod_i \text{Mat}_{n_i}(\text{End}_A(E_i)).$$

The integers n_i are multiplicities, and $\sum_i n_i$ is the length of the module.

1.3 Semisimple modules

Theorem. *For a module E , the following are equivalent: it is a sum of simple submodules; it is a direct sum of simple submodules; every submodule has a complement. Such a module is called semisimple.*

Proof. For a sum of simples, choose a maximal independent subfamily. Any simple summand not contained in their sum intersects it trivially, contradicting maximality; thus the direct sum is all of E . The same maximality argument, choosing a sum disjoint from a given submodule, produces a complement.

Proof. Conversely suppose all submodules split. Every nonzero cyclic submodule Av has a maximal proper submodule, corresponding to a maximal left ideal above the kernel of $A \rightarrow Av$. Splitting it exhibits a simple submodule of Av . Let F be the sum of all simple submodules of E . If $E = F \oplus F'$ with $F' \neq 0$, a cyclic submodule of F' contains a simple submodule, contradicting the definition of F . $\square$

Submodules and quotients of semisimple modules are semisimple. For a submodule, take its sum of simples and split it in E ; any residual nonzero complement inside the submodule would contain a simple. A quotient is isomorphic to a complement and is therefore semisimple as well.

1.4 The density theorem

For a module E , its commutant is $D = \text{End}_A(E)$ and its bicommutant is $\text{End}_D(E)$. Scalar multiplication gives a ring map $A \rightarrow \text{End}_D(E)$ because it commutes with all A -linear maps.

Theorem (Jacobson density). *If E is semisimple, $f \in \text{End}_D(E)$, and $x_1, \dots, x_n \in E$, there exists $a \in A$ with $ax_i = f(x_i)$ for all i . If E is finitely generated over D , the bicommutant map is surjective.*

Proof. First, for one x , split $E = Ax \oplus F$ and let π be the projection onto Ax . Since f commutes with π , $f(x) = f\pi(x) = \pi f(x) \in Ax$.

Proof. Apply this to E^n and the diagonal map f^n . The ring $\text{End}_A(E^n)$ is the matrix ring over D , so f^n commutes with it. The one-vector case applied to $(x_1, \dots, x_n)$ proves the assertion. Agreement on a finite D -generating set proves surjectivity. $\square$

The module is faithful if $\text{Ann}_A(E) = 0$. Thus, for simple faithful E finite-dimensional over D , the map $A \rightarrow \text{End}_D(E)$ is an isomorphism (Wedderburn's theorem in this form).

1.5 Burnside's theorem and projection operators

Theorem (Burnside). *Over an algebraically closed field k , if a subalgebra $A \subseteq \text{End}_k(E)$ acts simply on a finite-dimensional space E , then $A = \text{End}_k(E)$.*

Proof. $D = \text{End}_A(E)$ is a finite-dimensional division algebra with k central. For $d \in D$, the finite field extension $k[d]/k$ must equal k , so $D = k$. Density gives $A = \text{End}_D(E)$. $\square$

In particular, an irreducibly acting submonoid $G \subseteq \text{GL}(E)$ spans $\text{End}_k(E)$. A faithful irreducible finite-dimensional representation over algebraically closed k identifies its algebra with the full endomorphism algebra.

Proposition. For any field k , let $V_1, \dots, V_m$ be pairwise nonisomorphic simple A -modules finite-dimensional over k . There exist $e_i \in A$ acting as id on V_i and zero on the other V_j . *Proof.* the projections of $\bigoplus_i V_i$ commute with its commutant by Schur's lemma; apply density.

Proposition. If $\text{char } k = 0$, two finite-dimensional semisimple A -modules with the same trace function are isomorphic. The trace of e_i is the multiplicity of V_i times $\dim_k V_i$, so equality of traces gives equality of multiplicities. The same applies to semisimple monoid representations whose traces agree on all monoid elements, since these span the monoid algebra.

1.6 Semisimple rings and their components

A ring is semisimple if its left regular module is semisimple. In this discussion a simple ring means a semisimple ring whose simple left ideals are all isomorphic. Every module over a semisimple ring is semisimple: it is a quotient of a free module.

Proposition. If $L \subseteq A$ is a simple left ideal and E a simple module, then $LE = 0$ or $L \simeq E$. In the latter case choose y with $Ly \neq 0$; the map $L \rightarrow E$, $x \mapsto xy$, is an isomorphism.

Proof. Suppose $A = \bigoplus_{i=1}^s A_i$ as left ideals, with $A_i A_j = 0$ for $i \neq j$. Write $1 = \sum e_i$ with $e_i \in A_i$. Then $e_i e_j = \delta_{ij} e_i$, $ae_i = e_i a = a$ for $a \in A_i$, and each A_i is a two-sided ideal and ring with identity e_i . Thus $A \simeq \prod_i A_i$ as rings.

Theorem. A semisimple ring has finitely many simple-module isomorphism classes $L_1, \dots, L_s$. Its isotypic components

$$A_i = \sum_{L \subseteq A, L \simeq L_i} L$$

satisfy $A = \prod_i A_i$, with each A_i simple. Every module decomposes as $E = \bigoplus_i e_i E$, where $e_i E$ is the sum of its simple submodules isomorphic to L_i .

For the proof, express 1 using finitely many simple summands. These generate A , so only finitely many classes occur. The preceding lemma gives the vanishing of products of distinct components and identifies the simple constituents of $A_i E$. In particular, every simple module occurs as a simple left ideal; over a simple ring all simple modules are isomorphic.

1.7 Simple rings and Wedderburn's theorem

Proposition. Every endomorphism of the left module A is right multiplication: $\psi(x) = x\psi(1)$. A simple ring is a finite direct sum of simple left ideals, because finitely many summands suffice to contain 1.

Proposition. If L, M are simple left ideals, extend an isomorphism $L \rightarrow M$ by zero on a complement. It has the form $x \mapsto xa$, so $La = M$. Consequently $LA = A$. A nonzero two-sided ideal contains a simple left ideal and hence equals A . Every simple A -module is faithful and isomorphic to L .

Proof. Conversely, if a ring has no two-sided ideals except 0, A and has a nonzero simple left ideal L , then

$$A \xrightarrow{\sim} \text{End}_{\text{End}_A(L)}(L).$$

Indeed $LA = A$, so the regular module is a sum of images of L and is semisimple. The preceding results and density apply. More generally any nonzero left ideal of such a simple ring is a generator, and its bicommutant recovers A .

Proposition. If D is a division ring and E a finite-dimensional left D -space, then $A = \text{End}_D(E)$ is simple, E is a simple A -module, and $\text{End}_A(E) = D$ via left scalar multiplication. Given $v \neq 0$, a D -linear map can send v to any w , proving simplicity. Evaluation at a basis identifies $A \simeq E^n$ as A -modules. The density argument identifies the commutant.

For a field k and $\dim_k E = n$, $\dim_k \text{End}_k(E) = n^2$ and its left regular module has exactly n simple constituents, counted with multiplicity.

1.8 The Jacobson radical and Nakayama's lemma

Maximal left ideals correspond to simple cyclic modules with a chosen nonzero generator: $\mathfrak{m} \mapsto (A/\mathfrak{m}, 1)$ and $(E, x) \mapsto \text{Ann}_A(x)$. Define

$$J = \text{Rad}(A) = \bigcap_{\mathfrak{m} \text{ maximal left}} \mathfrak{m}.$$

Equivalently J is the intersection of the annihilators of all simple modules, hence is two-sided and annihilates every simple module.

Lemma (Nakayama). *If M is finitely generated and $JM = M$, then $M = 0$.*

Proof. Choose a minimal generating list $x_1, \dots, x_n$. Write $x_n = \sum a_i x_i$ with $a_i \in J$. Since $1 - a_n$ lies in no maximal left ideal, it has a left inverse b . Multiplication by b expresses x_n in terms of the other generators, a contradiction unless $n = 0$. $\square$

Every nilpotent two-sided ideal I lies in J : for a simple E , IE is either zero or E , and the second possibility contradicts $I^r = 0$.

For a finite-dimensional k -algebra, A/J is semisimple and J is nilpotent. Indeed a finite intersection of maximal left ideals equals J (successively decrease dimension), giving an embedding $A/J \hookrightarrow \bigoplus A/\mathfrak{m}_i$. Also J^r stabilizes, and Nakayama then gives $J^r = 0$. Thus $J = 0$ if and only if A is semisimple; in the reverse direction the faithful regular module is a sum of simples.

1.9 Semisimplicity under extension of scalars

Proposition. If A is a finite-dimensional semisimple k -algebra and K/k is finite separable, then $K \otimes_k A$ is semisimple.

Proof. To prove this, pass to a finite Galois closure; nonzero nilpotent ideals remain nonzero after field extension. Let K/k now be Galois and $N = \text{Rad}(K \otimes A)$. It is nilpotent and stable under the Galois group. If $0 \neq a = \sum x_i \otimes a_i \in N$ for a k -basis (a_i) , then for $y \in K$,

$$\sum_{\sigma} \sigma(ya) = \sum_i \text{Tr}_{K/k}(yx_i)a_i \in N \cap A.$$

The nondegenerate trace pairing gives a nonzero such sum, contradicting $N \cap A \subseteq \text{Rad}(A) = 0$.

Call A absolutely semisimple if $\bar{k} \otimes_k A$ is semisimple. Absolute semisimplicity implies semisimplicity because the radical, being nilpotent, extends into the radical after scalar extension. Finite-dimensional semisimple algebras over perfect fields are absolutely semisimple: any element of the extended radical and its finitely many coefficients descend to a finite separable extension, where no nonzero nilpotent ideal exists.

Over algebraically closed k , finite-dimensional simple algebras are $\text{End}_k(V)$, and

$$\text{End}_k(V) \otimes_k \text{End}_k(W) \simeq \text{End}_k(V \otimes_k W), \quad f \otimes g \mapsto (v \otimes w \mapsto f(v) \otimes g(w)).$$

Tensor products of simple algebras are therefore simple, and tensor products of semisimple algebras are semisimple by their product decompositions.

1.10 Balanced modules and generators

Write $A'(E) = \text{End}_A(E)$, acting on the left of E by evaluation, and $A''(E) = \text{End}_{A'(E)}(E)$. Thus all module structures in the statement below are left-module structures; equivalently one may use the customary opposite-ring formulation with right modules. A module is balanced if the natural map $A \rightarrow A''(E)$ is an isomorphism. It is a generator if every module is a quotient of a direct sum of copies of E .

Proposition. For every F , the module $A \oplus F$ is balanced. An element of its bicommutant commutes with the two projections and with all maps $A \rightarrow F$, $x \mapsto xu$. Its value at $(1, 0)$ is $(a, 0)$, and commutation forces its value at (x, u) to be (ax, au) .

Theorem (Morita). *E is a generator if and only if it is balanced and finitely generated projective over $A'(E)$.*

Proof. A generator admits a finite surjection $E^n \twoheadrightarrow A$, which splits, so $E^n \simeq A \oplus F$. The preceding lemma makes E^n balanced. An element of $A''(E)$ acts diagonally and commutes with all matrices over $A'(E)$, hence is scalar multiplication by some $a \in A$. Faithfulness follows from the surjection. Further,

$$A'(E)^n \simeq \text{Hom}_A(E^n, E) \simeq E \oplus \text{Hom}_A(F, E),$$

showing projectivity and finite generation over $A'(E)$.

Proof. Conversely, if $E \oplus F \simeq A'(E)^n$, applying $\text{Hom}_{A'(E)}(-, E)$ gives

$$E^n \simeq A''(E) \oplus \text{Hom}_{A'(E)}(F, E) \simeq A \oplus \text{Hom}_{A'(E)}(F, E).$$

Thus A is a quotient of E^n , so E generates. □

A nonzero left ideal L in a simple ring is an example: $LA = A$ gives a surjection $L^n \twoheadrightarrow A$.

2 Group representations

2.1 Representations and their constructions

Fix a commutative ring A and a group G . The group algebra $A[G]$ is free on the symbols $g \in G$, with multiplication extended from G . A G -module over A is an A -module E with an action $\rho : G \rightarrow \text{Aut}_A(E)$. Equivalently it is an $A[G]$ -module, or an algebra map $A[G] \rightarrow \text{End}_A(E)$. Morphisms are equivariant A -linear maps.

The representation is trivial if every g acts as the identity, and faithful if $\ker \rho = \{1\}$. Faithfulness as a group representation does not imply faithfulness as an $A[G]$ -module.

The standard constructions carry actions

$$\begin{aligned} g(x, y) &= (gx, gy) && \text{on } E \oplus F, \\ g(x \otimes y) &= gx \otimes gy && \text{on } E \otimes_A F, \\ (gf)(x) &= g(f(g^{-1}x)) && \text{on } \text{Hom}_A(E, F). \end{aligned}$$

Use the trivial action on A . The dual is $E^\vee = \text{Hom}_A(E, A)$, with matrix $\rho^\vee(g) = \rho(g^{-1})^\top$. For finite free E , the map

$$E^\vee \otimes_A M \xrightarrow{\sim} \text{Hom}_A(E, M), \quad f \otimes y \mapsto (x \mapsto f(x)y)$$

is equivariant.

The invariants are $E^G = \{x : gx = x \text{ for all } g\}$. The invariants functor is right adjoint to giving an A -module the trivial action, and is left exact. In particular

$$\text{Hom}_A(E, F)^G = \text{Hom}_{A[G]}(E, F).$$

2.2 Averaging and Maschke's theorem

Let G be finite of order n . The averaging trace on any G -module is

$$\mathrm{Tr}_G(x) = \sum_{g \in G} gx \in E^G.$$

It is natural in E .

$$\begin{array}{ccc} E & \xrightarrow{\varphi} & F \\ \mathrm{Tr}_G \downarrow & & \downarrow \mathrm{Tr}_G \\ E^G & \xrightarrow{\varphi^G} & F^G \end{array}$$

Applied to Hom modules, averaging commutes with precomposition and postcomposition by equivariant maps $u : E' \rightarrow E$ and $v : F \rightarrow F'$.

$$\begin{array}{ccc} \mathrm{Hom}_k(E, F) & \xrightarrow{v \circ (-) \circ u} & \mathrm{Hom}_k(E', F') \\ \mathrm{Tr}_G \downarrow & & \downarrow \mathrm{Tr}_G \\ \mathrm{Hom}_{k[G]}(E, F) & \xrightarrow{v \circ (-) \circ u} & \mathrm{Hom}_{k[G]}(E', F') \end{array}$$

Theorem (Maschke). *If k is a field and $\mathrm{char} k \nmid n$, then $k[G]$ is semisimple.*

Proof. For a submodule $F \subseteq E$, choose a k -linear retraction $p : E \rightarrow F$. The map $n^{-1} \mathrm{Tr}_G(p)$ is equivariant and still restricts to the identity on F . Thus every submodule splits. $\square$

Let $E_1, \dots, E_s$ represent $\mathrm{Irr}_k(G)$. Every finite-dimensional representation is uniquely $\bigoplus E_i^{m_i}$. There are central orthogonal idempotents e_i acting as identity on E_i and zero on the others, and

$$k[G] = \prod_i A_i, \quad A_i = k[G]e_i \simeq \mathrm{End}_{D_i}(E_i), \quad D_i = \mathrm{End}_{k[G]}(E_i).$$

Every module decomposes into the isotypic summands $e_i E$.

Proposition. If k is algebraically closed, $D_i = k$; writing $d_i = \dim_k E_i$ gives

$$k[G] \simeq \prod_i \mathrm{Mat}_{d_i}(k), \quad k[G] \simeq \bigoplus_i E_i^{d_i}, \quad n = \sum_i d_i^2.$$

The center is $\oplus_i ke_i$.

2.3 Characters and multiplicities

For finite-dimensional E , its character is $\chi_E(g) = \text{Tr}(\rho_E(g))$, extended linearly to $k[G]$. Isomorphic representations have equal characters. The trivial character is $1_G(g) = 1$, and

$$\chi_{E \oplus F} = \chi_E + \chi_F, \quad \chi_{E \otimes F} = \chi_E \chi_F, \quad \chi_{E^\vee}(g) = \chi_E(g^{-1}).$$

An effective character is the character of a representation. A character, in the manuscript's broader convention, is an integral linear combination of effective characters (often called a virtual character). A simple or irreducible character comes from a simple module.

Assume $\text{char } k \nmid |G|$. There are finitely many simple characters, and every effective character is a nonnegative integral combination of them. In characteristic zero this expression is unique, and

$$E \simeq F \iff \chi_E = \chi_F.$$

Indeed evaluate a relation $\sum a_i \chi_i = 0$ on the central projection e_j : it gives $a_j d_j = 0$. Thus the simple characters are linearly independent, and traces determine multiplicities.

The dimension of an effective character is $\dim \chi_E = \dim_k E$; its multiplicity of χ_i is m_i in $\chi_E = \sum m_i \chi_i$. Dimension is additive under direct sums, hence defines a homomorphism of additive monoids. The manuscript postpones the positive-characteristic independence statement until the central-idempotent formula.

2.4 One-dimensional characters and the center

One-dimensional representations correspond to group homomorphisms $G \rightarrow k^\times$. They form a group under pointwise multiplication. For $n = |G|$, all values lie in $\mu_n(k) = \{z : z^n = 1\}$.

Proposition. If G is abelian, k is algebraically closed, and $\text{char } k \nmid n$, all simple representations are one-dimensional: the commutative algebra $k[G] \simeq \prod_i \text{Mat}_{d_i}(k)$ forces $d_i = 1$. Restricting a character of any finite G to $\langle g \rangle$ shows that $\chi(g)$ is an integral linear combination of n th roots of unity. Let $\text{Conj}(G)$ denote the conjugacy classes, and put $c_C = \sum_{g \in C} g$. Then

$$Z(k[G]) = \bigoplus_{C \in \text{Conj}(G)} k c_C.$$

Indeed $\sum a_g g$ is central precisely when its coefficients are constant on conjugacy classes, by conjugating with each group element. Comparing with $\bigoplus_i k e_i$ gives

$$|\text{Irr}_k(G)| = |\text{Conj}(G)|.$$

The regular representation is the left action on $k[G]$. Its character is

$$\chi_{\text{reg}} = \sum_i d_i \chi_i, \quad \chi_{\text{reg}}(g) = \begin{cases} n & g = 1, \\ 0 & g \neq 1. \end{cases}$$

In the group-element basis, left multiplication by a nonidentity element has no fixed basis vector, giving trace zero.

2.5 Central idempotents and class functions

Throughout this subsection and the ensuing orthogonality discussion, k is algebraically closed and $n = |G|$ is invertible in k . The primitive central idempotents have the explicit form

$$e_i = \frac{d_i}{n} \sum_{g \in G} \chi_i(g^{-1})g.$$

For $e_i = \sum a_g g$, the coefficient a_t is $n^{-1} \chi_{\text{reg}}(e_i t^{-1}) = n^{-1} d_i \chi_i(t^{-1})$. Thus $d_i \neq 0$ in k , since $e_i \neq 0$, and the simple characters are linearly independent even in this characteristic.

Their coefficients lie in the subring $\mathbb{Z}[\mu_n, 1/n]$ of k . In characteristic zero each d_i divides n . To see this, the formula implies that multiplication by $(n/d_i)e_i$ preserves the finitely generated $\mathbb{Z}[\mu_n]$ -module spanned by the ge_i . The determinant argument shows that n/d_i is an algebraic integer; being rational, it is an integer.

Definition. A class function is a function constant on conjugacy classes. Write $X_k(G)$ for their space and extend its elements linearly to $k[G]$. Characters are class functions because trace is invariant under conjugation. The pairing

$$X_k(G) \times Z(k[G]) \rightarrow k, \quad (f, z) \mapsto f(z)$$

is perfect: on class indicators and class sums it has diagonal entries $|C|$, all invertible. In the other bases,

$$\chi_i(e_j) = \delta_{ij} d_i.$$

Consequently the irreducible characters form a basis of $X_k(G)$.

2.6 Orthogonality of matrix coefficients

For functions on G , define

$$\langle f, h \rangle = \frac{1}{n} \sum_{g \in G} f(g)h(g^{-1}).$$

With convolution $(f * h)(t) = \sum_g f(g)h(g^{-1}t)$ this is $n^{-1}(f * h)(1)$.

For $x \in E$ and $\lambda \in E^\vee$, the coefficient function is $c_{E,\lambda,x}(g) = \lambda(gx)$. In a basis and its dual these are the matrix entries of $\rho_E(g)$.

For nonisomorphic simple E, F , all pairings of their coefficients vanish. Indeed averaging the rank-one map $E \rightarrow F$, $v \mapsto \lambda(v)y$, gives an equivariant map, hence zero by Schur's lemma.

For a simple E of dimension d one has

$$\langle c_{E,\lambda,x}, c_{E,\mu,y} \rangle = \frac{1}{d} \lambda(y) \mu(x).$$

To prove this, average $T(v) = \lambda(v)y$ using the Hom action. The result is a scalar $a \operatorname{id}_E$ by Schur's lemma. Since $\operatorname{Tr}(T) = \lambda(y)$ and conjugation preserves trace,

$$da = n\lambda(y).$$

Evaluate the averaged map at x and apply μ , then divide by n . The factor d is invertible by the central-idempotent formula.

The identity $\operatorname{Tr}(v \mapsto \lambda(v)y) = \lambda(y)$ follows directly from a basis containing y (or from the coordinate matrix).

2.7 Character orthogonality and unitary representations

Summing diagonal matrix coefficients gives

$$\langle \chi_i, \chi_j \rangle = \delta_{ij}, \quad f = \sum_i \langle f, \chi_i \rangle \chi_i \quad (f \in X_k(G)).$$

The latter follows by expanding in the character basis and pairing with each χ_j .

Now take $k = \mathbb{C}$. Use the Hermitian pairing

$$(f, h) = \frac{1}{n} \sum_g f(g) \overline{h(g)}.$$

For characters, $\chi(g^{-1}) = \overline{\chi(g)}$, since eigenvalues are roots of unity. Thus the bilinear and Hermitian pairings agree on characters, and $\chi_{E^\vee} = \overline{\chi_E}$.

Proposition. Every finite-dimensional complex representation admits an invariant positive definite Hermitian inner product: average any such product,

$$(x, y)_G = \sum_g (gx, gy)_0.$$

In an orthonormal basis its matrices are unitary, so $\rho(g^{-1}) = \overline{\rho(g)}^\top$.

Proof. Choose unitary matrix models $\rho^{(i)}$ for the irreducibles. Their coefficient functions satisfy

$$(\rho_{ab}^{(i)}, \rho_{cd}^{(j)}) = \frac{\delta_{ij} \delta_{ac} \delta_{bd}}{d_i}.$$

There are $\sum_i d_i^2 = n$ such nonzero orthogonal functions, so they form a basis of all functions $G \rightarrow \mathbb{C}$. The corresponding Fourier expansion is

$$f = \sum_i d_i \sum_{a,b=1}^{d_i} (f, \rho_{ab}^{(i)}) \rho_{ab}^{(i)}.$$

2.8 Irreducibility tests and permutation characters

Proposition. An effective complex character $\chi = \sum_i m_i \chi_i$ is irreducible exactly when

$$(\chi, \chi) = \sum_i m_i^2 = 1, \quad \text{equivalently} \quad \sum_{g \in G} |\chi(g)|^2 = |G|.$$

For finite-dimensional complex representations,

$$(\chi_E, \chi_F) = \dim_{\mathbb{C}} \operatorname{Hom}_G(E, F).$$

This follows either from the multiplicity decompositions or from $\operatorname{Hom}_{\mathbb{C}}(E, F) \simeq E^{\vee} \otimes F$: the dimension of its invariant subspace is the multiplicity of the trivial character, obtained by averaging.

Proposition. For a permutation action on a finite set S , the permutation character is

$$\chi(g) = |S^g|.$$

Its average is the number of G -orbits on S (count pairs (g, s) with $gs = s$). For the natural action of S_n on $\{1, \dots, n\}$,

$$\sum_{g \in S_n} \chi(g) = n!.$$

More generally (χ, χ) is the number of orbits on $S \times S$. Thus a doubly transitive permutation action has permutation character $1_G + \theta$ with θ irreducible, since its norm is two and its trivial multiplicity is one.

2.9 Bimodules, tensor products, and adjunction

An (A, B) -bimodule has commuting left A - and right B -actions: $(am)b = a(mb)$. Morphisms respect both. For a right A -module M and left A -module N , $M \otimes_A N$ represents biadditive balanced maps, satisfying $(ma, n) \mapsto (m, an)$.

It can be constructed as the cokernel of the difference of the two maps

$$M \otimes_{\mathbb{Z}} A \otimes_{\mathbb{Z}} N \rightrightarrows M \otimes_{\mathbb{Z}} N,$$

sending $m \otimes a \otimes n$ to $ma \otimes n$ and $m \otimes an$. If M is an (A, B) -bimodule and N a (B, C) -bimodule, then $M \otimes_B N$ is an (A, C) -bimodule.

For a ring map $A \rightarrow B$, extension of scalars is left adjoint to restriction:

$$\mathrm{Hom}_B(B \otimes_A M, N) \simeq \mathrm{Hom}_A(M, N|_A).$$

The bijection sends f to $m \mapsto f(1 \otimes m)$; its inverse sends u to $b \otimes m \mapsto bu(m)$. The balanced relation is exactly the condition that this inverse be well-defined. Equivalently, applying Hom to the cokernel construction gives this adjunction by a kernel computation.

For $M' \rightarrow M \rightarrow M'' \rightarrow 0$ exact, adjunction identifies the Hom sequences below, proving right exactness of extension of scalars:

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathrm{Hom}_B(B \otimes_A M'', N) & \rightarrow & \mathrm{Hom}_B(B \otimes_A M, N) & \rightarrow & \mathrm{Hom}_B(B \otimes_A M', N) \\ & & \sim \downarrow & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & \mathrm{Hom}_A(M'', N_A) & \longrightarrow & \mathrm{Hom}_A(M, N_A) & \longrightarrow & \mathrm{Hom}_A(M', N_A) \end{array}$$

2.10 Induction and Frobenius reciprocity

For $H \leq G$, set

$$\mathrm{Res}_H^G E = E|_H, \quad \mathrm{Ind}_H^G F = k[G] \otimes_{k[H]} F.$$

The tensor adjunction gives Frobenius reciprocity:

$$\mathrm{Hom}_G(\mathrm{Ind}_H^G F, E) \simeq \mathrm{Hom}_H(F, \mathrm{Res}_H^G E).$$

If R represents the left cosets G/H , then $k[G] = \bigoplus_{r \in R} rk[H]$ as a right module, hence $\mathrm{Ind}_H^G F = \bigoplus_{r \in R} r \otimes F$. In particular an H -submodule $F \subseteq E$ induces E via the natural map exactly when $E = \bigoplus_{r \in R} rF$.

Induction is additive and exact, and is transitive:

$$\mathrm{Ind}_H^G \mathrm{Ind}_K^H \simeq \mathrm{Ind}_K^G.$$

The regular representation is $\mathrm{Ind}_{\{1\}}^G k$. The tensor identity is

$$\mathrm{Ind}_H^G(\mathrm{Res}_H^G E \otimes_k F) \simeq E \otimes_k \mathrm{Ind}_H^G F,$$

via $g \otimes (x \otimes y) \mapsto gx \otimes (g \otimes y)$; the coset decompositions show it is bijective.

2.11 Induced characters

Let G be finite and work over an algebraically closed field of characteristic not dividing $|G|$. For $E = \text{Ind}_H^G F$,

$$\begin{aligned}\chi_E(g) &= \sum_{\substack{r \in R \\ r^{-1}gr \in H}} \chi_F(r^{-1}gr) \\ &= \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1}gx \in H}} \chi_F(x^{-1}gx).\end{aligned}$$

Indeed g permutes the coset summands rF , and only fixed summands contribute to its trace. On a fixed summand its action is conjugate to $r^{-1}gr$ on F .

The same formula defines $\text{Ind}_H^G f$ for a class function f on H ; restriction is $\text{Res}_H^G h = h|_H$. They satisfy

$$\langle \text{Ind}_H^G f, h \rangle_G = \langle f, \text{Res}_H^G h \rangle_H.$$

This follows directly by changing variables in the sums, or in characteristic zero by Frobenius reciprocity and the character formula for Hom dimensions, then linearity. The direct sum computation also establishes the formula in the other allowed characteristics.

2.12 Mackey's decomposition and irreducibility criterion

Let $H, K \leq G$ and choose representatives S for $K \backslash G / H$. Put $H_s = sHs^{-1} \cap K$, and view F as an H_s -module F_s via $h \mapsto \rho(s^{-1}hs)$. Then

$$\operatorname{Res}_K^G \operatorname{Ind}_H^G F \simeq \bigoplus_{s \in S} \operatorname{Ind}_{H_s}^K F_s.$$

For the proof, split the induced space into spans of the summands xF with $x \in KsH$. Each is a K -module induced from sF , whose stabilizer is H_s ; the action on sF identifies it with F_s .

For complex representations (or under the same splitting-field hypotheses), $\operatorname{Ind}_H^G F$ is irreducible if and only if F is irreducible and

$$\operatorname{Hom}_{H_s}(F|_{H_s}, F_s) = 0 \quad (s \in G \setminus H),$$

where now $K = H$. Indeed reciprocity and the decomposition express its endomorphism dimension as the sum of these intertwining dimensions. The identity double coset contributes $\dim \operatorname{End}_H(F)$, and all terms are nonnegative integers.

Proposition. If $H \triangleleft G$, this simplifies to: F is irreducible and $F_s \not\cong F$ for every $s \notin H$.

2.13 Cyclic groups and direct products

Over algebraically closed k with $|G|$ invertible, G is abelian if and only if every irreducible has dimension one, by the matrix-factor decomposition of $k[G]$.

For $C_n = \langle r \rangle$, choose a primitive n th root ζ . Its irreducible characters are

$$\chi_h(r^j) = \zeta^{hj} \quad (h, j \in \mathbb{Z}/n\mathbb{Z}).$$

The character group is cyclic, generated by χ_1 . A character table has one row per irreducible character and one column per conjugacy class, with entry $\chi_i(C_j)$.

For $G_1 \times G_2$, the irreducibles are exactly the external tensor products $E_i \boxtimes F_j$, with

$$(g, h)(x \otimes y) = gx \otimes hy, \quad \chi_{E_i \boxtimes F_j}(g, h) = \chi_{E_i}(g)\chi_{F_j}(h).$$

Their character norms factor and equal one, so they are irreducible and pairwise distinct. They exhaust the irreducibles since the sum of their squared dimensions is

$$\left(\sum_i \dim(E_i)^2\right)\left(\sum_j \dim(F_j)^2\right) = |G_1||G_2|.$$

If $A \leq G$ is abelian, each irreducible E has $\dim E \leq [G : A]$: choose a one-dimensional constituent λ of $E|_A$, use reciprocity to make E a quotient of $\text{Ind}_A^G \lambda$, and compare dimensions.

2.14 Dihedral groups

Write $D_n = \langle r, s : r^n = s^2 = 1, srs = r^{-1} \rangle$, of order $2n$. The cyclic subgroup $\langle r \rangle$ has index two, so irreducibles have degree at most two. Let ζ be a primitive n th root. Inducing $\chi_h(r) = \zeta^h$ gives

$$\rho_h(r) = \begin{pmatrix} \zeta^h & 0 \\ 0 & \zeta^{-h} \end{pmatrix}, \quad \rho_h(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Mackey's criterion says this is irreducible unless $h \equiv -h \pmod{n}$, and $\rho_h \simeq \rho_{n-h}$. Its character on r^j is $\zeta^{hj} + \zeta^{-hj}$ and on reflections is zero.

Proposition. If n is even, the classes are $\{1\}$, $\{r^{n/2}\}$, the pairs $\{r^j, r^{-j}\}$ for $1 \leq j < n/2$, and two reflection classes distinguished by parity in sr^j . There are four linear characters

$$\chi_{ab}(r) = (-1)^a, \quad \chi_{ab}(s) = (-1)^b \quad (a, b \in \{0, 1\}),$$

and the degree-two characters ρ_h , $1 \leq h < n/2$. Their squared degrees sum to $4 + 4(n/2 - 1) = 2n$.

Proposition. If n is odd, the rotation classes are $\{1\}$ and the pairs $\{r^j, r^{-j}\}$ for $1 \leq j \leq (n-1)/2$, and all reflections form one class. There are two linear characters, $r \mapsto 1$, $s \mapsto \pm 1$, and $(n-1)/2$ degree-two characters ρ_h .

2.15 Character tables of the alternating and symmetric groups

For $A_4 = V_4 \rtimes C_3$, inflation gives three linear characters. Inducing a nontrivial character of V_4 gives an irreducible of degree three, because its three conjugates are distinct. With $x \in V_4 \setminus \{1\}$, a three-cycle t , and $\omega^3 = 1$, $\omega \neq 1$, the table is

	1	x	t	t^2
$ C $	1	3	4	4
1	1	1	1	1
χ_1	1	1	ω	ω^2
χ_2	1	1	ω^2	ω
ψ	3	-1	0	0

For S_4 , inflation along $S_4/V_4 \simeq S_3$ gives degrees 1, 1, 2. The natural permutation representation is $1 \oplus \theta$, where θ is irreducible of degree three by double transitivity. Its sign twist gives the remaining row:

	1	(12)	(123)	(1234)	(12)(34)
$ C $	1	6	8	6	3
1	1	1	1	1	1
ε	1	-1	1	-1	1
η	2	0	-1	0	2
θ	3	1	0	-1	-1
$\varepsilon\theta$	3	-1	0	1	-1

Here θ is the action on $\{(a_1, a_2, a_3, a_4) : \sum a_i = 0\}$; its character is the number of fixed letters minus one.

2.16 Column orthogonality and $\mathrm{SL}(2,3)$

Character orthogonality also gives the column formula

$$\sum_i \chi_i(g) \chi_i(h^{-1}) = \begin{cases} |G|/|g^G| & g, h \text{ conjugate,} \\ 0 & \text{otherwise.} \end{cases}$$

Expand the indicator of a conjugacy class in the irreducible-character basis to obtain it.

The action of $\mathrm{SL}_2(\mathbb{F}_3)$ on $\mathbb{P}^1(\mathbb{F}_3)$ has kernel $\{\pm I\}$ and image A_4 . This group has order 24 and seven conjugacy classes. Choose x of order four and t of order three. Four irreducibles are inflated from A_4 ; the remaining squared degrees sum to 12, so they all have degree two. With ω as above the completed table is

	1	x	t	t^2	$-I$	$-t$	$-t^2$
$ C $	1	6	4	4	1	4	4
1	1	1	1	1	1	1	1
χ_1	1	1	ω	ω^2	1	ω	ω^2
χ_2	1	1	ω^2	ω	1	ω^2	ω
ψ	3	-1	0	0	3	0	0
ρ	2	0	-1	-1	-2	1	1
$\chi_1\rho$	2	0	$-\omega$	$-\omega^2$	-2	ω	ω^2
$\chi_2\rho$	2	0	$-\omega^2$	$-\omega$	-2	ω^2	ω

Column orthogonality forces the new characters to be -2 at $-I$ and zero at x . Twisting by χ_1, χ_2 gives the other degree-two rows; orthogonality and the root-of-unity values determine the entries at t .

2.17 Artin's induction theorem

Work over $\mathbb{C}$. Let $X(G) = \bigoplus_{\chi \in \text{Irr}(G)} \mathbb{Z}\chi$ be the character ring, and $X_R(G) = R \otimes_{\mathbb{Z}} X(G)$. Restriction is a ring map; induction is additive and satisfies

$$\langle \text{Ind } f, g \rangle_G = \langle f, \text{Res } g \rangle_H, \quad \text{Ind}(f \text{ Res } g) = (\text{Ind } f)g.$$

Thus the image of induction from any family of subgroups is an ideal.

Theorem (Artin). *For a family $\mathcal{F}$ of subgroups of G , the following are equivalent:*

1. *Every element of G is conjugate to an element of some $H \in \mathcal{F}$.*

2. $\bigoplus_{H \in \mathcal{F}} X_{\mathbb{Q}}(H) \xrightarrow{\sum \text{Ind}_H^G} X_{\mathbb{Q}}(G)$ *is surjective.*

Proof. Condition (1) is equivalent to injectivity of the joint restriction map on class functions. The induction and restriction maps are adjoint under the nondegenerate character pairings, so injectivity of restriction is equivalent to surjectivity of induction. Their matrices on character lattices have integer entries, so the rank statement holds over $\mathbb{Q}$ as well as $\mathbb{C}$. $\square$

Taking all cyclic subgroups yields: every character is a rational linear combination of characters induced from cyclic subgroups.

2.18 Brauer's induction theorem

Fix a prime p . Every $g \in G$ has a unique commuting factorization $g = g_u g_r$, where g_u has p -power order and g_r has order prime to p ; both lie in $\langle g \rangle$. These are its p -unipotent and p -regular parts.

A p -elementary subgroup has the form $C \times P$, with C cyclic of order prime to p and P a p -group. Let

$$I^p(G) = \sum_{H \text{ } p\text{-elementary}} \text{Ind}_H^G X(H) \subseteq X(G).$$

This is an ideal. If $|G| = p^a \ell$ with $(p, \ell) = 1$, Brauer's key assertion is $\ell 1_G \in I^p(G)$. It implies the finite quotient $X(G)/I^p(G)$ is annihilated by ℓ (its index therefore divides a power of ℓ).

Varying p makes these integers relatively prime, so

$$X(G) = \sum_p I^p(G).$$

Every character is therefore an integral linear combination of characters induced from elementary subgroups. The manuscript also records that irreducible characters of elementary groups are monomial: induced from degree-one characters of subgroups. By transitivity of induction this yields Brauer's theorem in its usual form:

Theorem (Brauer). *Every complex character of a finite group is an integral linear combination of monomial characters.*

The following page retains the manuscript's proof strategy for the key assertion.

2.19 Brauer induction: the integral construction

Proof. Put $n = |G| = p^a \ell$ and $\mathcal{O} = \mathbb{Z}[\zeta_n]$. First, for an integer-valued class function f , one has $nf \in \mathcal{O}I^p(G)$. For each cyclic H , let $\theta_H(h) = |H|$ if h generates H , and zero otherwise. Directly counting conjugates gives

$$n 1_G = \sum_{H \leq G \text{ cyclic}} \text{Ind}_H^G \theta_H.$$

The Fourier coefficients of $\theta_H f|_H$ are sums of integers times roots of unity, hence lie in $\mathcal{O}$. The tensor identity proves the assertion, since cyclic groups are p -elementary.

Next construct an integer-valued $\psi \in \mathcal{O}I^p(G)$ whose values are all prime to p . For a p -regular s , put $C = \langle s \rangle$, choose $P \in \text{Syl}_p(C_G(s))$, and set $H = C \times P$. On C take the function $\psi_C = |C| 1_{\{s\}}$ and inflate it to H . Its Fourier coefficients lie in $\mathcal{O}$. Its induction ψ_s is integer-valued, is zero on other p -regular conjugacy classes, and has

$$\psi_s(s) = |C_G(s)|/|P| \not\equiv 0 \pmod{p}.$$

Sum these functions over the p -regular classes. For any integer-valued $f \in X_{\mathcal{O}}(G)$, the root-of-unity expansion and a sufficiently large p -power give $f(g) \equiv f(g_r) \pmod{p}$. Thus the sum ψ is prime to p everywhere.

Proof. Choose N with $\psi(g)^N \equiv 1 \pmod{p^a}$ for all g . Applying the first step to $(\psi^N - 1)/p^a$ gives $\ell(\psi^N - 1) \in \mathcal{O}I^p(G)$. Since $\psi \in \mathcal{O}I^p(G)$, it follows that $\ell \in \mathcal{O}I^p(G)$. Projecting the free $\mathbb{Z}$ -module $\mathcal{O}$ onto its coefficient of 1 gives $\ell \in I^p(G)$.

3 Homological algebra

3.1 Additive and abelian categories

A category is preadditive if every Hom set is an abelian group and composition is bilinear. A functor is additive if its maps on Hom groups are homomorphisms. An object is initial if and only if it is terminal if and only if its identity morphism is zero; such an object is a zero object.

In a preadditive category, a finite product, when it exists, is also a coproduct, denoted $X \oplus Y$. Its inclusions and projections satisfy

$$p_i \iota_j = \delta_{ij}, \quad \iota_X p_X + \iota_Y p_Y = \text{id}.$$

Additive functors preserve zero objects and these biproducts. A category is additive if it is preadditive and has all finite biproducts, including a zero object.

For $f : X \rightarrow Y$, define

$$\ker f = \text{eq}(f, 0), \quad \text{coker } f = \text{coeq}(f, 0),$$

$$\text{Im } f = \ker(\text{coker } f), \quad \text{coim } f = \text{coker}(\ker f).$$

Kernels and image inclusions are monic; cokernels and coimage projections are epic. The universal properties give a canonical factorization

$$X \twoheadrightarrow \text{coim } f \longrightarrow \text{Im } f \hookrightarrow Y.$$

An abelian category is additive, has all kernels and cokernels, and makes the middle map an isomorphism for every f .

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \uparrow \\ \text{coim } f & \xrightarrow{\sim} & \text{Im } f \end{array}$$

Example. Examples are abelian groups, left or right modules over a ring, the opposite of an abelian category, and functor categories $\mathcal{A}^I$ for small I

and abelian $\mathcal{A}$. In a functor category kernels and cokernels are computed objectwise.

3.2 Chain complexes and homology

Let $\mathcal{A}$ be abelian and $I \subseteq \mathbb{Z}$ an interval. A chain complex has objects X_n and differentials $d_n : X_n \rightarrow X_{n-1}$ with $d_{n-1}d_n = 0$. A chain map $f : X \rightarrow Y$ satisfies $d_Y f_n = f_{n-1}d_X$. Write $\text{Ch}_I(\mathcal{A})$ for the resulting category; $\text{Ch} = \text{Ch}_{\mathbb{Z}}$ and $\text{Ch}_+ = \text{Ch}_{\mathbb{Z}_{\geq 0}}$.

$$\begin{array}{ccccc}
 X_{n+1} & \xrightarrow{d_X} & X_n & \xrightarrow{d_X} & X_{n-1} \\
 f_{n+1} \downarrow & & \downarrow f_n & & \downarrow f_{n-1} \\
 Y_{n+1} & \xrightarrow{d_Y} & Y_n & \xrightarrow{d_Y} & Y_{n-1}
 \end{array}$$

This category is abelian. Sums, kernels, and cokernels are computed degree by degree; the induced differentials still square to zero. Extending a complex by zero gives the natural full embeddings for nested intervals.

Definition. A sequence $X' \xrightarrow{f} X \xrightarrow{g} X''$ is exact at X if $\text{Im } f = \ker g$. The cycles, boundaries, and homology of a complex are

$$Z_n(X) = \ker d_n, \quad B_n(X) = \text{Im } d_{n+1}, \quad H_n(X) = Z_n(X)/B_n(X).$$

Chain maps preserve cycles and boundaries and induce homology maps, making Z_n, B_n, H_n additive functors.

Definition. A quasi-isomorphism is a chain map inducing isomorphisms on all homology objects. A complex is exact, or acyclic, if all homology vanishes; equivalently $0 \rightarrow X$ is a quasi-isomorphism.

3.3 Exact functors and the embedding theorem

Definition. An additive functor $F : \mathcal{A} \rightarrow \mathcal{B}$ is left exact if every short exact sequence $0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$ gives an exact sequence

$$0 \rightarrow F(X') \rightarrow F(X) \rightarrow F(X'').$$

It is right exact if the resulting sequence

$$F(X') \rightarrow F(X) \rightarrow F(X'') \rightarrow 0$$

is exact, and exact if both hold.

The functors $\text{Hom}(X, -) : \mathcal{A} \rightarrow \mathbf{Ab}$ and $\text{Hom}(-, X) : \mathcal{A}^{\text{op}} \rightarrow \mathbf{Ab}$ are left exact. This is the universal property of kernels, including the reversed-arrow formulation for the contravariant Hom functor.

For an adjoint pair of additive functors $F \dashv G$ between abelian categories, F is right exact and G left exact. Indeed left adjoints preserve cokernels and right adjoints preserve kernels.

Theorem (Freyd–Mitchell). *Every small abelian category admits a fully faithful exact embedding into the category of left modules over a ring.*

The notes use this theorem to justify proving finite diagram lemmas by element arguments for modules, and then applying them in an abelian category.

3.4 Double complexes and totalization

Definition. A double complex consists of $X_{p,q}$ with horizontal and vertical differentials

$$d^h : X_{p,q} \rightarrow X_{p-1,q}, \quad d^v : X_{p,q} \rightarrow X_{p,q-1},$$

satisfying $(d^h)^2 = (d^v)^2 = 0$ and $d^h d^v + d^v d^h = 0$. Morphisms commute with both. The sign change $\partial_{p,q}^v = (-1)^p d_{p,q}^v$ turns the anticommuting squares into commuting ones, identifying double complexes with chain complexes of chain complexes.

$$\begin{array}{ccc} X_{p,q} & \xrightarrow{d^h} & X_{p-1,q} \\ \partial^v \downarrow & & \downarrow \partial^v \\ X_{p,q-1} & \xrightarrow{d^h} & X_{p-1,q-1} \end{array}$$

The product and sum total complexes are

$$\mathrm{Tot}_n^\Pi X = \prod_{p+q=n} X_{p,q}, \quad \mathrm{Tot}_n^\oplus X = \bigoplus_{p+q=n} X_{p,q},$$

with differential $d^h + d^v$. They exist when the required products or coproducts exist. The manuscript calls a double complex bounded when each diagonal $p + q = n$ has only finitely many nonzero terms. In that case both totalizations agree.

A diagonally finite double complex with exact rows, or with exact columns, has acyclic total complex. A cycle can be eliminated successively along a diagonal: solve using exactness, subtract a total boundary, and repeat; diagonal finiteness makes the process terminate.

3.5 Truncations, shifts, and mapping cones

The good truncation is

$$(\tau_{\geq n} X)_k = \begin{cases} 0 & k < n, \\ Z_n(X) & k = n, \\ X_k & k > n, \end{cases} \quad \tau_{< n} X = X / \tau_{\geq n} X.$$

Its homology agrees with $H_k(X)$ for $k \geq n$ and is zero below n ; the quotient has the complementary homology.

Use the manuscript's homological shift convention:

$$X[p]_k = X_{p+k}, \quad d_{X[p]} = (-1)^p d_X, \quad H_k(X[p]) = H_{p+k}(X).$$

Viewing $Z(X), B(X), H(X)$ as complexes with zero differential, one obtains

$$0 \rightarrow Z(X) \rightarrow X \xrightarrow{d} B(X)[-1] \rightarrow 0,$$

and

$$0 \rightarrow H(X) \rightarrow X/B(X) \xrightarrow{d} Z(X)[-1] \rightarrow H(X)[-1] \rightarrow 0.$$

For a chain map $f : X \rightarrow Y$, its mapping cone is the total complex

$$\text{Cone}(f)_n = Y_n \oplus X_{n-1}, \quad d(y, x) = (d_Y y + f(x), -d_X x).$$

The underlying chain-map diagram is:

$$\begin{array}{ccccc} X_n & \xrightarrow{d_X} & X_{n-1} & \xrightarrow{d_X} & X_{n-2} \\ f_n \downarrow & & \downarrow f_{n-1} & & \downarrow f_{n-2} \\ Y_n & \xrightarrow{d_Y} & Y_{n-1} & \xrightarrow{d_Y} & Y_{n-2} \end{array}$$

The signs make $d^2 = 0$, and there is a natural short exact sequence

$$0 \rightarrow Y \rightarrow \text{Cone}(f) \rightarrow X[-1] \rightarrow 0.$$

3.6 The snake lemma and the homology sequence

For a commutative diagram with exact rows $X \rightarrow Y \rightarrow Z \rightarrow 0$ and $0 \rightarrow X' \rightarrow Y' \rightarrow Z'$, and vertical maps α, β, γ , the diagram is

$$\begin{array}{ccccccc} X & \longrightarrow & Y & \longrightarrow & Z & \longrightarrow & 0 \\ & & \alpha \downarrow & & \downarrow \beta & & \downarrow \gamma \\ 0 & \longrightarrow & X' & \longrightarrow & Y' & \longrightarrow & Z' \end{array}$$

Theorem. The snake lemma gives

$$\ker \alpha \rightarrow \ker \beta \rightarrow \ker \gamma \xrightarrow{\delta} \operatorname{coker} \alpha \rightarrow \operatorname{coker} \beta \rightarrow \operatorname{coker} \gamma.$$

For modules, lift $z \in \ker \gamma$ to $y \in Y$. Then $\beta(y)$ comes from $x' \in X'$, and $\delta(z)$ is the class of x' modulo $\operatorname{Im} \alpha$. Independence of choices, exactness, and naturality follow by this element chase.

Proposition. Every short exact sequence of complexes $0 \rightarrow X \rightarrow Y \rightarrow Z \rightarrow 0$ induces a natural long exact sequence

$$\cdots \rightarrow H_n(X) \rightarrow H_n(Y) \rightarrow H_n(Z) \xrightarrow{\partial} H_{n-1}(X) \rightarrow \cdots$$

To compute $\partial[z]$, lift the cycle z to $y \in Y_n$; then $d_Y y$ is the image of a cycle $x \in X_{n-1}$, and $\partial[z] = [x]$. This is the snake construction applied to the relevant cycles and boundaries.

$$\begin{array}{ccccccc} X_n/B_n(X) & \longrightarrow & Y_n/B_n(Y) & \longrightarrow & Z_n/B_n(Z) & \longrightarrow & 0 \\ & & \bar{d}_X \downarrow & & \downarrow \bar{d}_Y & & \downarrow \bar{d}_Z \\ 0 & \longrightarrow & Z_{n-1}(X) & \longrightarrow & Z_{n-1}(Y) & \longrightarrow & Z_{n-1}(Z) \end{array}$$

The construction is functorial in maps of short exact sequences, giving a functor from the category of short exact sequences of complexes to long exact sequences in $\mathcal{A}$.

3.7 Split complexes and chain homotopies

Definition. A complex is split if there are maps $s_n : C_n \rightarrow C_{n+1}$ with $d = dsd$. It is split exact if it is also acyclic. For example the doubly infinite exact complex with every term $\mathbb{Z}/4$ and every differential multiplication by 2 is not split: $2s2 = 0$ for every endomorphism s .

Proposition. A split complex has decompositions

$$C_n \simeq B_n \oplus H_n \oplus B_{n-1},$$

where the differential is the identity from the last summand onto the first summand in degree $n - 1$, and zero elsewhere. *Proof.* ds splits off B_n in Z_n , while s splits the surjection $C_n \rightarrow B_{n-1}$. Conversely such decompositions give a splitting s . Projection onto and inclusion of the H_n summands give quasi-isomorphisms $C \rightleftarrows H(C)$, with zero differential on $H(C)$.

Proposition. A bounded-below acyclic complex of projective modules is split exact: starting at its lowest degree, split $0 \rightarrow Z_n \rightarrow C_n \rightarrow B_{n-1} \rightarrow 0$ inductively; each boundary is then projective.

For chain maps $f, g : C \rightarrow D$, a chain homotopy is a family $s_n : C_n \rightarrow D_{n+1}$ such that

$$f - g = d_D s + s d_C.$$

$$\begin{array}{ccccc}
 C_{n+1} & \xrightarrow{d_C} & C_n & \xrightarrow{d_C} & C_{n-1} \\
 \downarrow f_{n+1} - g_{n+1} & \swarrow s_n & \downarrow f_n - g_n & \swarrow s_{n-1} & \downarrow f_{n-1} - g_{n-1} \\
 D_{n+1} & \xrightarrow{d_D} & D_n & \xrightarrow{d_D} & D_{n-1}
 \end{array}$$

Definition. A map homotopic to zero is null-homotopic. A complex is split exact precisely when its identity is null-homotopic. A chain homotopy equivalence has a chain-map inverse up to homotopy. Homotopic maps induce equal homology maps: on a cycle, their difference is a boundary. Thus a homotopy equivalence is a quasi-isomorphism.

3.8 Cones, cylinders, and homotopy equivalences

The cone of id_C is contractible: with our coordinates $(y, x) \in C_n \oplus C_{n-1}$, the contraction is $s(y, x) = (0, y)$. A chain map $f : C \rightarrow D$ is null-homotopic precisely when it extends across $C \hookrightarrow \text{Cone}(\text{id}_C)$; a homotopy $f = ds + sd$ gives the extension $(y, x) \mapsto f(y) + s(x)$.

$$\begin{array}{ccc}
 C & \xrightarrow{y \mapsto (y, 0)} & \text{Cone}(\text{id}_C) \\
 & \searrow f & \downarrow (y, x) \mapsto f(y) + s(x) \\
 & & D
 \end{array}$$

The long exact sequence of $0 \rightarrow C \rightarrow \text{Cone}(f) \rightarrow B[-1] \rightarrow 0$, for $f : B \rightarrow C$, has connecting map f_* under the shift identification. Therefore

$$f \text{ is a quasi-isomorphism} \iff \text{Cone}(f) \text{ is acyclic.}$$

Define the cylinder by

$$\text{Cyl}(f)_n = B_n \oplus B_{n-1} \oplus C_n,$$

$$d(b, u, c) = (db + u, -du, dc - fu).$$

The chain maps $\alpha(c) = (0, 0, c)$ and $\beta(b, u, c) = f(b) + c$ satisfy $\beta\alpha = \text{id}_C$ and

$$\text{id} - \alpha\beta = ds + sd, \quad s(b, u, c) = (0, b, 0).$$

Thus the cylinder is homotopy equivalent to C . For $f = \text{id}_C$, two maps $C \rightarrow D$ are homotopic precisely when they extend together to a map from this cylinder.

3.9 Short exact sequences and their cones

The cylinder construction factors any map $f : B \rightarrow C$ as a monomorphism $B \rightarrow \text{Cyl}(f)$ followed by the homotopy equivalence $\beta : \text{Cyl}(f) \rightarrow C$. Here $j(b) = (b, 0, 0)$. Its quotient by B is a cone of f , up to the indicated coordinate sign change.

$$\begin{array}{ccc} B & \xrightarrow{j} & \text{Cyl}(f) \\ & \searrow f & \downarrow \beta \\ & & C \end{array}$$

Also

$$0 \rightarrow C \rightarrow \text{Cyl}(f) \rightarrow \text{Cone}(-\text{id}_B) \rightarrow 0,$$

whose quotient is contractible.

Proof. Suppose $0 \rightarrow B \xrightarrow{f} C \xrightarrow{g} D \rightarrow 0$ is a short exact sequence of complexes. Then

$$q : \text{Cone}(f) \rightarrow D, \quad q(c, b) = g(c),$$

is a quasi-isomorphism. Compare the cone homology sequence with the original homology sequence: the maps on $H(B)$ and $H(C)$ are identities, and the five lemma gives that q_* is an isomorphism.

$$\begin{array}{ccccccc} H_n(B) & \xrightarrow{f_*} & H_n(C) & \rightarrow & H_n(\text{Cone}(f)) & \xrightarrow{-p_*} & H_{n-1}(B) \\ \parallel & & \parallel & & \downarrow q_* & & \parallel \\ H_n(B) & \xrightarrow{f_*} & H_n(C) & \xrightarrow{g_*} & H_n(D) & \xrightarrow{\partial} & H_{n-1}(B) \end{array}$$

Here $p(c, b) = b$; the minus sign follows from $dc = -f(b)$ for a cone cycle. Equivalently, the kernel of q is a contractible cone on B .

These comparisons recover the original connecting maps with the indicated signs.

Similarly $B[-1] \rightarrow \text{Cone}(g)$ is a quasi-isomorphism, given in our coordinates by $b \mapsto (0, f(b))$. The cone sequence therefore packages the same homology information as the short exact sequence.

3.10 Abelian subcategories and presheaves

Definition. An abelian subcategory has an exact inclusion into the ambient abelian category. A full subcategory is additive when it contains a zero object and is closed under finite direct sums; it is an abelian subcategory if it is also closed under ambient kernels and cokernels.

Finitely generated modules form an additive subcategory, and an abelian subcategory over a left noetherian ring. Torsion-free abelian groups form an additive category; finite abelian p -groups and $\mathbb{F}_p$ -vector spaces give full abelian subcategories of $\mathbf{Ab}$.

Graded abelian groups form $\mathbf{Ab}^{\mathbb{Z}}$ with $\mathbb{Z}$ discrete. Chain complexes form a full abelian subcategory of the appropriate diagram category. Viewing a ring as a one-object preadditive category identifies modules with additive functors to abelian groups.

For a space X , a presheaf with values in $\mathcal{A}$ is a functor $\mathrm{Op}(X)^{\mathrm{op}} \rightarrow \mathcal{A}$. An abelian-group-valued presheaf F is a sheaf if for every open cover $U = \bigcup_i U_i$,

$$0 \rightarrow F(U) \rightarrow \prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j)$$

is an equalizer (equivalently, use the difference of the two restriction maps).

Example. Examples are smooth real-valued functions on a manifold and the constant sheaf A_X , whose sections are locally constant maps $U \rightarrow A$ with A discrete. Sheaves form a full subcategory of presheaves, but the inclusion is not exact.

3.11 The exponential sequence and Yoneda's embedding

Let $\mathcal{O}_X$ and $\mathcal{O}_X^\times$ be the sheaves of continuous complex-valued and nonvanishing complex-valued functions. There is an exact sequence of sheaves of abelian groups

$$0 \rightarrow \mathbb{Z}_X \xrightarrow{2\pi i} \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^\times \rightarrow 1.$$

The kernel statement is pointwise. Surjectivity as a sheaf map follows from local logarithms: cover $\mathbb{C}^\times$ by domains admitting logarithms, and pull the cover back along a section. Any sheaf morphism annihilating the image of $\exp$ annihilates a section locally and hence globally.

For $X = \mathbb{C}^\times$ this sequence is not exact as presheaves: the identity function has no continuous logarithm on all of X , so the map on global sections is not surjective. Sheaf inclusion has sheafification as a left adjoint and is therefore left exact.

Yoneda sends $X \in \mathcal{A}$ to $h_X = \text{Hom}_{\mathcal{A}}(-, X)$ and is fully faithful and left exact as a functor into additive presheaves. Its values are left exact functors. For small abelian $\mathcal{A}$, the category of left exact additive functors $\mathcal{A}^{\text{op}} \rightarrow \mathbf{Ab}$ gives an abelian enlargement in which Yoneda is exact. This is part of the embedding-theorem discussion in the manuscript.

3.12 Cochain complexes and delta functors

A cochain complex has $d^n : X^n \rightarrow X^{n+1}$ and $d^{n+1}d^n = 0$. It is a reindexed chain complex, with $X^n = X_{-n}$ and $d^n = d_{-n}$. Write Z^n, B^n, H^n for its cocycles, coboundaries, and cohomology. A short exact sequence gives

$$\cdots \rightarrow H^n(B) \rightarrow H^n(C) \rightarrow H^n(D) \xrightarrow{\partial} H^{n+1}(B) \rightarrow \cdots .$$

A homological δ -functor consists of additive $T_n : \mathcal{A} \rightarrow \mathcal{B}$ for $n \geq 0$ and natural connecting maps $T_n(C) \rightarrow T_{n-1}(A)$ for every short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, forming long exact sequences. A cohomological δ -functor has T^n and connecting maps $T^n(C) \rightarrow T^{n+1}(A)$. Set negative degrees to zero. Necessarily T_0 is right exact and T^0 left exact. Homology and cohomology are the basic examples.

$$\begin{array}{ccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\ & & \downarrow a & & \downarrow b & & \downarrow c \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' \longrightarrow 0 \end{array}$$

$$\begin{array}{ccc} T_n(C) & \xrightarrow{\partial} & T_{n-1}(A) \\ T_n(c) \downarrow & & \downarrow T_{n-1}(a) \\ T_n(C') & \xrightarrow{\partial} & T_{n-1}(A') \end{array}$$

Proposition. For a central element a of a ring, the snake lemma applied to multiplication by a gives the example

$$T_0(M) = M/aM, \quad T_1(M) = M[a] = \{m : am = 0\}, \quad T_n = 0 \quad (n > 1).$$

A short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ gives

$$0 \rightarrow M'[a] \rightarrow M[a] \rightarrow M''[a] \rightarrow M'/aM' \rightarrow M/aM \rightarrow M''/aM'' \rightarrow 0.$$

3.13 Universal delta functors and resolutions

A morphism of δ -functors is a family of natural transformations commuting with the connecting morphisms. A homological $T_\bullet$ is universal if every natural map $S_0 \rightarrow T_0$ from another homological δ -functor extends uniquely to a morphism $S_\bullet \rightarrow T_\bullet$. Dually, a cohomological $T^\bullet$ is universal if every map $T^0 \rightarrow S^0$ extends uniquely to all degrees.

For an exact additive functor F , setting $T_0 = F$ and all higher terms zero gives a universal δ -functor in either sense. A universal extension of a right exact F is unique up to unique isomorphism and its higher functors are called left satellites; the cohomological version gives right satellites of a left exact functor.

Definition. A left resolution of X is a quasi-isomorphism $P_\bullet \rightarrow X$ with $P_n = 0$ for $n < 0$, viewing X in degree zero. Equivalently

$$\cdots \rightarrow P_2 \rightarrow P_1 \rightarrow P_0 \rightarrow X \rightarrow 0$$

is exact. A right resolution is a quasi-isomorphism $X \rightarrow I^\bullet$ in nonnegative cohomological degrees, or an exact sequence

$$0 \rightarrow X \rightarrow I^0 \rightarrow I^1 \rightarrow I^2 \rightarrow \cdots .$$

3.14 Projective objects and projective resolutions

For an object P of an abelian category, these are equivalent:

1. Every map $P \rightarrow C$ lifts across every epimorphism $B \twoheadrightarrow C$.
2. $\text{Hom}(P, -)$ is exact.
3. Every short exact sequence ending in P splits.

Such an object is projective. Left exactness of Hom makes (1) equivalent to (2). For (3) to imply (1), pull back $B \twoheadrightarrow C$ along $P \rightarrow C$. The resulting epimorphism $B \times_C P \rightarrow P$ splits, and projection to B gives the desired lift. Conversely apply the lifting property to id_P . The pullback square, with $q : B \twoheadrightarrow C$, sits in:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \ker q & \longrightarrow & B \times_C P & \twoheadrightarrow & P \longrightarrow 0 \\
 & & \parallel & & \downarrow & & \downarrow f \\
 0 & \longrightarrow & \ker q & \longrightarrow & B & \xrightarrow{q} & C \longrightarrow 0
 \end{array}$$

A module is projective if and only if it is a direct summand of a free module. A category has enough projectives if every object is a quotient of a projective. Module categories have this property.

Definition. A projective resolution is a left resolution whose terms are projective. When there are enough projectives, construct one inductively: choose $P_0 \twoheadrightarrow X$, then $P_1 \twoheadrightarrow \ker(P_0 \rightarrow X)$, then a projective surjection onto each successive kernel. Composing these surjections with the kernel inclusions gives the differentials and ensures exactness.

3.15 The comparison theorem

Theorem (Comparison). *Let $f : M \rightarrow N$, let $P_\bullet$ be a nonnegative complex of projectives augmented to M , and let $Q_\bullet \rightarrow N$ be a resolution. Then f lifts to a chain map $P_\bullet \rightarrow Q_\bullet$, uniquely up to chain homotopy.*

Lift in degree zero using $Q_0 \twoheadrightarrow N$. If the lift is defined through degree n , its composite with d_{n+1} lands in $Z_n(Q) = \text{Im } d_{n+1}^Q$; projectivity of P_{n+1} supplies the next lift. At degree zero, use $\ker(Q_0 \rightarrow N)$ in place of $Z_0(Q)$.

$$\begin{array}{ccccc}
 P_{n+1} & \xrightarrow{\quad d_P \quad} & P_n & & \\
 \downarrow f_{n+1} & \searrow f_n d_P & \downarrow f_n & & \\
 Q_{n+1} & \twoheadrightarrow Z_n(Q) & \hookrightarrow Q_n & &
 \end{array}$$

Proof. For uniqueness let $h = f_\bullet - g_\bullet$. Inductively construct s with $h = ds + sd$. In degree zero, h_0 maps into the augmentation kernel, so lifts to Q_1 . Given the previous homotopy terms, $h_n - s_{n-1}d$ is a cycle-valued map and lifts to Q_{n+1} . In particular any two projective resolutions of an object are homotopy equivalent.

3.16 The horseshoe lemma

Lemma (Horseshoe). *Given $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ and projective resolutions $P' \rightarrow A'$, $P'' \rightarrow A''$, there is a projective resolution $P \rightarrow A$ and a short exact sequence of complexes*

$$0 \rightarrow P' \rightarrow P \rightarrow P'' \rightarrow 0, \quad P_n = P'_n \oplus P''_n,$$

lifting the given sequence, with the natural inclusions and projections.

$$\begin{array}{ccccccc}
 & \vdots & & \vdots & & \vdots & \\
 & \downarrow & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & P'_1 & \longrightarrow & P_1 & \longrightarrow & P''_1 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & P'_0 & \longrightarrow & P_0 & \longrightarrow & P''_0 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & A'' \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

Construct the augmentation by lifting $P''_0 \rightarrow A''$ to A . The snake lemma gives an exact sequence of augmentation kernels. Repeat on these kernels, lifting the next projective term at each stage, to construct the differentials.

3.17 Injective objects and Baer's criterion

Definition. An object I is injective if maps into I extend across monomorphisms, equivalently $\text{Hom}(-, I)$ is exact, or I is projective in the opposite category. A category has enough injectives if every object embeds into one. Repeatedly embedding the successive cokernels constructs an injective resolution.

$$\begin{array}{ccccc}
 0 & \longrightarrow & X & \xhookrightarrow{j} & X' \\
 & & \downarrow f & \nwarrow \tilde{f} & \\
 & & I & &
 \end{array}$$

Theorem (Baer's criterion). *A left A -module I is injective if and only if every map from a left ideal $J \subseteq A$ into I extends to A .*

A module over a commutative ring is divisible if multiplication by each non-zero-divisor is surjective. Injective modules are divisible, by applying Hom to $0 \rightarrow A \xrightarrow{a} A \rightarrow A/aA \rightarrow 0$. Over a PID the converse follows from Baer's criterion, since every ideal is principal. In particular $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are injective abelian groups.

Abelian groups have enough injectives. Products of injectives are injective, and the evaluation map

$$M \hookrightarrow I_0(M) = (\mathbb{Q}/\mathbb{Z})^{\text{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})}$$

is injective: for $x \neq 0$, choose a character of $\langle x \rangle$ nonzero on x and extend it to M using injectivity of $\mathbb{Q}/\mathbb{Z}$.

For an additive adjunction $F \dashv G$, exactness of G makes F preserve projectives, while exactness of F makes G preserve injectives. This follows by applying the Hom adjunction to the defining exactness property.

3.18 Enough injectives for modules and sheaves

The forgetful functor from left A -modules to abelian groups is exact and has right adjoint $J \mapsto \operatorname{Hom}_{\mathbb{Z}}(A, J)$, with $(af)(b) = f(ba)$. Thus $\operatorname{Hom}_{\mathbb{Z}}(A, J)$ is injective if J is. For an embedding of underlying groups $e : M \hookrightarrow I_0(M)$, the map

$$M \hookrightarrow \operatorname{Hom}_{\mathbb{Z}}(A, I_0(M)), \quad m \mapsto (a \mapsto e(am))$$

is an injective A -linear map. Hence module categories have enough injectives.

For a sheaf F on X , its stalk at x is $F_x = \varinjlim_{U \ni x} F(U)$. Directed colimits of abelian groups are exact, so stalk functors are exact. The stalk functor is left adjoint to the skyscraper functor

$$(x_*A)(U) = \begin{cases} A & x \in U, \\ 0 & x \notin U. \end{cases}$$

Indeed $\operatorname{Hom}(F, x_*A) \simeq \operatorname{Hom}(F_x, A)$ by the colimit universal property. Thus x_* carries injective groups to injective sheaves.

Embed each stalk F_x into an injective group I_x . The natural map

$$F \hookrightarrow \prod_{x \in X} x_*I_x$$

is monic: a section with every germ zero is locally zero and hence zero. The target is injective, proving that sheaves of abelian groups have enough injectives and therefore admit injective resolutions.

3.19 Constructing left derived functors

Let $F : \mathcal{A} \rightarrow \mathcal{B}$ be right exact and additive, with $\mathcal{A}$ having enough projectives. Choose a projective resolution $P \rightarrow A$ and define

$$L_i F(A) = H_i(F(P)).$$

Right exactness gives $L_0 F \simeq F$. Comparison maps between two resolutions are inverse up to homotopy. Additive functors preserve homotopies, so the induced homology isomorphisms are canonical and independent of choices.

For a map $f : A \rightarrow A'$, choose a comparison lift $\tilde{f} : P \rightarrow P'$ and set $L_i F(f) = H_i(F(\tilde{f}))$. Homotopy uniqueness makes this independent of the lift. Lifts of identities, composites, and sums prove functoriality and additivity, with $L_0 F(f) = F(f)$.

$$\begin{array}{ccc} P_{\bullet} & \xrightarrow{\tilde{f}} & P'_{\bullet} \\ \downarrow & & \downarrow \\ A & \xrightarrow{f} & A' \end{array}$$

Projective objects have $L_i F(P) = 0$ for $i > 0$. An object Q with this vanishing is called F -acyclic. An F -acyclic resolution has all terms F -acyclic; it will also compute $L_i F(A)$.

Proposition. If $U : \mathcal{B} \rightarrow \mathcal{C}$ is exact, then

$$U \circ L_i F \simeq L_i(U \circ F),$$

because exact functors commute with homology. In particular, forgetful functors to abelian groups often allow a derived-functor calculation to be checked on the underlying groups.

3.20 Exact sequences and naturality

The left derived functors form a homological δ -functor. For $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$, use the horseshoe lemma to obtain a degreewise split sequence of projective resolutions. Applying the additive F preserves these splittings, so the homology sequence gives

$$\cdots \rightarrow L_n F(A') \rightarrow L_n F(A) \rightarrow L_n F(A'') \xrightarrow{\partial} L_{n-1} F(A') \rightarrow \cdots.$$

To verify naturality, lift a map of short exact sequences to their horseshoe resolutions. In the decompositions $P_n = P'_n \oplus P''_n$, write

$$d_P = \begin{pmatrix} d' & \lambda_P \\ 0 & d'' \end{pmatrix}, \quad \tilde{f}_n = \begin{pmatrix} f'_n & \gamma_n \\ 0 & f''_n \end{pmatrix}.$$

The required off-diagonal relation is

$$d'_Q \gamma_n = \gamma_{n-1} d''_P + f'_{n-1} \lambda_P - \lambda_Q f''_n.$$

The right side is cycle-valued by the previously constructed relations and $d^2 = 0$, so projectivity supplies γ_n . Naturality of the homology sequence then applies.

$$\begin{array}{ccccccc} 0 & \longrightarrow & P'_\bullet & \longrightarrow & P_\bullet & \longrightarrow & P''_\bullet \longrightarrow 0 \\ & & f'_\bullet \downarrow & & \downarrow \tilde{f}_\bullet & & \downarrow f''_\bullet \\ 0 & \longrightarrow & Q'_\bullet & \longrightarrow & Q_\bullet & \longrightarrow & Q''_\bullet \longrightarrow 0 \end{array}$$

For a map from the sequence on A', A, A'' to one on B', B, B'' :

$$\begin{array}{ccc} L_n F(A'') & \xrightarrow{\partial} & L_{n-1} F(A') \\ L_n F(f'') \downarrow & & \downarrow L_{n-1} F(f') \\ L_n F(B'') & \xrightarrow{\partial} & L_{n-1} F(B') \end{array}$$

3.21 Dimension shifting and acyclic resolutions

Proposition. If $0 \rightarrow M \rightarrow P \rightarrow A \rightarrow 0$ has P F -acyclic, dimension shifting gives

$$L_i F(A) \simeq L_{i-1} F(M) \quad (i \geq 2), \quad L_1 F(A) \simeq \ker(F(M) \rightarrow F(P)).$$

Iterating through an F -acyclic resolution $Q \rightarrow A$ proves

$$L_i F(A) \simeq H_i(F(Q)).$$

Thus projectivity of every term is sufficient but not necessary for computation.

3.22 Universality and right derived functors

The δ -functor $L_\bullet F$ is universal. Given $T_\bullet$ and $T_0 \rightarrow F$, choose $0 \rightarrow K \rightarrow P \rightarrow A \rightarrow 0$ with P projective. Since $L_n F(P) = 0$ for $n > 0$, the map $L_n F(A) \rightarrow L_{n-1} F(K)$ is monic. By induction, the composite

$$T_n(A) \xrightarrow{\partial} T_{n-1}(K) \rightarrow L_{n-1} F(K)$$

lands in its image, defining $T_n(A) \rightarrow L_n F(A)$ uniquely. Comparison diagrams prove independence of P , naturality, and compatibility with all connecting maps. The same monomorphism proves uniqueness inductively.

$$\begin{array}{ccccc} T_n(A) & \xrightarrow{\partial} & T_{n-1}(K) & \longrightarrow & T_{n-1}(P) \\ \theta_n \downarrow & & \downarrow \theta_{n-1} & & \downarrow \theta_{n-1} \\ L_n F(A) & \xrightarrow{\partial} & L_{n-1} F(K) & \longrightarrow & L_{n-1} F(P) \end{array}$$

Dually, for a left exact additive F and enough injectives, choose $A \rightarrow I^\bullet$ and define

$$R^i F(A) = H^i(F(I^\bullet)).$$

This is a universal cohomological δ -functor with $R^0 F = F$. It vanishes in positive degrees on injectives. An object is F -acyclic if $R^i F = 0$ for $i > 0$; any resolution by such objects computes the derived functors.

For modules, define

$$\mathrm{Ext}_A^i(M, N) = R^i \mathrm{Hom}_A(M, -)(N).$$

The later balancing argument shows that projectively resolving M gives the same groups as injectively resolving N .

3.23 Ext criteria and sheaf cohomology

Proposition. For an A -module N , the following are equivalent: N is injective; $\text{Hom}_A(-, N)$ is exact; $\text{Ext}_A^i(M, N) = 0$ for all M and $i > 0$; $\text{Ext}_A^1(M, N) = 0$ for all M .

For the last implication, embed N into an injective I and write $Q = I/N$. The long exact sequence and $\text{Ext}_A^1(Q, N) = 0$ show $\text{Hom}(Q, I) \rightarrow \text{Hom}(Q, N)$ is onto. Lifting id_Q splits the embedding, so N is an injective direct summand.

Dually, M is projective if and only if $\text{Hom}_A(M, -)$ is exact, if and only if all $\text{Ext}_A^i(M, N)$ vanish for $i > 0$ and all N , if and only if all $\text{Ext}_A^1(M, N)$ vanish. Here the degree-one vanishing makes Hom preserve epimorphisms by its long exact sequence.

Proposition. For a space X , the global-sections functor $\Gamma(X, F) = F(X)$ on sheaves of abelian groups is left exact. Its right derived functors are sheaf cohomology:

$$H^i(X, F) = R^i\Gamma(X, -)(F).$$

Enough injectives for sheaves makes this definition available on any space. The exponential sequence is one source of the resulting long exact cohomology sequences.

3.24 Adjunctions, sums, and filtered colimits

Proposition. A left adjoint preserves all existing colimits, and a right adjoint preserves all existing limits. An indexing category is filtered if it is nonempty, every pair of objects has a common target, and every pair of parallel arrows becomes equal after some further arrow.

Filtered colimits of modules are exact. An abelian category satisfies AB5 if it is cocomplete and filtered colimits are exact. Module categories satisfy AB5; the opposite category of abelian groups does not.

For module-valued functors in this setting, derived functors of a left adjoint commute with direct sums and filtered colimits:

$$L_n F(\bigoplus_i X_i) \simeq \bigoplus_i L_n F(X_i), \quad L_n F(\varinjlim_i X_i) \simeq \varinjlim_i L_n F(X_i).$$

For sums, take sums of projective resolutions, apply the colimit-preserving F , and use exactness of sums to commute them with homology. For filtered colimits, use compatible projective resolutions of the diagram, followed by exactness of the filtered colimit and the same homology argument.

3.25 The acyclic assembly lemma

Let C be a double complex of modules. “Upper half-plane” means $C_{p,q} = 0$ for $q < 0$, and “right half-plane” means $C_{p,q} = 0$ for $p < 0$, after a possible translation.

Lemma (Acyclic assembly). *The product total complex is acyclic for an upper half-plane complex with exact columns, or a right half-plane complex with exact rows. The direct-sum total complex is acyclic for an upper half-plane complex with exact rows, or a right half-plane complex with exact columns.*

For the upper half-plane product case, translate to consider a cycle $c \in \text{Tot}_0^\Pi(C)$. Starting at $q = 0$, solve successively for a degree-one tuple b with $db = c$. Once earlier components are chosen, the remaining component to be lifted is a vertical cycle, because $dc = 0$ and the differentials anticommute. Exactness of its column supplies the next component. Products permit the resulting infinite tuple.

The direct-sum case follows by truncating in the other direction and using that a cycle has only finitely many nonzero components. This reduces to a diagonally finite complex, where product and sum totalizations agree. Interchanging the axes gives the other two cases.

Remark. The choice of totalization matters for unbounded diagonals: an infinite tuple may be a cycle in the product totalization without belonging to the direct sum. The manuscript’s example using repeated multiplication by 2 on $\mathbb{Z}/4$ illustrates this, with a product-total homology class represented by a constant infinite tuple, while the sum totalization is acyclic.

3.26 Tor and the tensor-product double complex

For a (B, A) -bimodule M , tensor–Hom adjunction gives

$$\mathrm{Hom}_B(M \otimes_A N, P) \simeq \mathrm{Hom}_A(N, \mathrm{Hom}_B(M, P)).$$

Thus $M \otimes_A -$ is right exact. For a right A -module M and left A -module N , define

$$\mathrm{Tor}_n^A(M, N) = L_n(M \otimes_A -)(N).$$

Tor commutes with direct sums and filtered colimits in N by the preceding results.

For chain complexes P of right modules and Q of left modules, their tensor double complex has

$$(P \otimes_A Q)_{p,q} = P_p \otimes_A Q_q, \quad d^h = d_P \otimes 1, \quad d^v = (-1)^p 1 \otimes d_Q.$$

The tensor-product complex is its direct-sum totalization.

Theorem (Balancing Tor). *Either variable may be projectively resolved:*

$$L_n(M \otimes_A -)(N) \simeq L_n(- \otimes_A N)(M).$$

Choose projective resolutions $P \rightarrow M$ and $Q \rightarrow N$. The augmentations give maps

$$M \otimes_A Q \longleftarrow \mathrm{Tot}(P \otimes_A Q) \longrightarrow P \otimes_A N.$$

Both are quasi-isomorphisms: their cones arise from augmented double complexes with exact rows or columns, since projective modules are flat. Acyclic assembly applies. Taking homology proves balancing, and hence commutation with sums and filtered colimits in the other variable too.

3.27 Balancing Ext

For a chain complex P and cochain complex I , set

$$\mathrm{Hom}(P, I)^{p,q} = \mathrm{Hom}_A(P_p, I^q).$$

One compatible differential convention is $d^h f = f \circ d_P$ and $d^v f = (-1)^{p+q+1} d_I \circ f$ for $f \in \mathrm{Hom}(P, I)^{p,q}$; the two anticommute. Its product totalization is the Hom cochain complex.

For chain complexes C, D , reindex D cohomologically. A degree- n cocycle in Hom is a family $C_p \rightarrow D_{p-n}$ satisfying the signed chain-map relation; a coboundary is a null-homotopy. Consequently

$$H^n \mathrm{Hom}(C, D) \simeq \{\text{chain maps } C \rightarrow D[-n]\} / \text{homotopy}.$$

Tensor–Hom adjunction extends to complexes, relating the Hom complex of a sum-total tensor complex to the product-total iterated Hom complex, with the corresponding signs.

Proof. Choose a projective resolution $P \rightarrow M$ and an injective resolution $N \rightarrow I$. Both maps

$$\mathrm{Hom}_A(M, I) \longrightarrow \mathrm{Tot} \mathrm{Hom}_A(P, I) \longleftarrow \mathrm{Hom}_A(P, N)$$

are quasi-isomorphisms, by exactness of Hom against projectives or injectives and the cochain version of acyclic assembly. Hence

$$\mathrm{Ext}_A^n(M, N) \simeq H^n \mathrm{Hom}_A(P, N) \simeq H^n \mathrm{Hom}_A(M, I).$$

This supplies the promised balance between the two variables.

3.28 Tor for abelian groups

For an integer $r > 0$, resolve $\mathbb{Z}/r$ by $0 \rightarrow \mathbb{Z} \xrightarrow{r} \mathbb{Z} \rightarrow \mathbb{Z}/r \rightarrow 0$. Tensoring with B gives

$$\mathrm{Tor}_0^{\mathbb{Z}}(\mathbb{Z}/r, B) = B/rB, \quad \mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/r, B) = B[r],$$

and all higher Tor groups vanish.

Proposition. For any abelian groups A, B , $\mathrm{Tor}_1^{\mathbb{Z}}(A, B)$ is torsion and $\mathrm{Tor}_n^{\mathbb{Z}}(A, B) = 0$ for $n \geq 2$. Express A as the filtered union of its finitely generated subgroups. Tor commutes with this colimit, and the structure theorem reduces each finitely generated subgroup to free summands and cyclic torsion summands, for which the assertion follows from the calculation above.

Since $\mathbb{Q}/\mathbb{Z}$ is the filtered union of the finite cyclic subgroups $\frac{1}{r}\mathbb{Z}/\mathbb{Z}$, the same computation gives

$$\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}, B) \simeq \bigcup_{r>0} B[r] = B_{\mathrm{tors}}.$$

The transition maps under this identification are inclusions of the respective torsion subgroups.

3.29 Further Tor computations

Proposition. A torsion-free abelian group is a filtered union of finitely generated free groups. Hence its higher Tor groups vanish. Conversely $\mathrm{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/r, A) = A[r]$, so vanishing of Tor_1 against every group characterizes torsion-freeness.

For $A = \mathbb{Z}/m$, $d \mid m$, and $e = m/d$, the periodic free resolution of $\mathbb{Z}/d$ alternates multiplication by d and e . Thus for an A -module B ,

$$\mathrm{Tor}_n^A(\mathbb{Z}/d, B) = \begin{cases} B/dB & n = 0, \\ B[d]/eB & n > 0 \text{ odd}, \\ B[e]/dB & n > 0 \text{ even}. \end{cases}$$

For a domain A with fraction field K , flatness of localization and $0 \rightarrow A \rightarrow K \rightarrow K/A \rightarrow 0$ give

$$\mathrm{Tor}_1^A(K/A, N) = \{x \in N : ax = 0 \text{ for some } a \neq 0\}.$$

If I is a right ideal and J a left ideal in a ring, then

$$\mathrm{Tor}_1^A(A/I, A/J) \simeq (I \cap J)/IJ.$$

Indeed dimension shifting identifies it with the kernel of $I \otimes_A A/J \rightarrow A \otimes_A A/J$. Since these are I/IJ and A/J , the kernel is as stated. In particular, for a two-sided ideal I , $\mathrm{Tor}_1^A(A/I, A/I) = I/I^2$.

For $A = k[x, y]$ and $I = (x, y)$, the Koszul resolution

$$0 \rightarrow A \xrightarrow{(-y, x)} A^2 \xrightarrow{(x, y)} A \rightarrow k \rightarrow 0$$

gives $\mathrm{Tor}_2^A(k, k) = k$ and $\mathrm{Tor}_1^A(k, k) = k^2$. Dimension shifting also gives $\mathrm{Tor}_1^A(k, I) = k$.

3.30 Flatness and localization

Definition. A right module M is flat if $M \otimes_A -$ is exact; a left module N is flat if $- \otimes_A N$ is exact. For a left module these are equivalent:

1. N is flat.
2. $\mathrm{Tor}_n^A(M, N) = 0$ for all right modules M and all $n > 0$.
3. $\mathrm{Tor}_1^A(M, N) = 0$ for every right module M .

Flatness makes a tensor-applied projective resolution exact in positive degrees. Conversely, degree-one vanishing and the Tor long exact sequence show that tensoring preserves monomorphisms, hence is exact.

Proposition. If $S \subseteq Z(A)$ is multiplicatively closed, then $S^{-1}A$ is flat on both sides. Regard it as a filtered colimit of copies of A , with transition maps multiplication by the relevant elements of S . Filtered colimits are exact, so a filtered colimit of flat modules is flat.

For $0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0$ with N'' flat, the Tor sequence shows

$$N' \text{ flat} \iff N \text{ flat}.$$

In particular all projective modules are flat, and torsion-free abelian groups are exactly the flat $\mathbb{Z}$ -modules.

3.31 Character duality and the ideal criterion for flatness

For a left module N , its character dual is the right module

$$N^* = \text{Hom}_{\mathbb{Z}}(N, \mathbb{Q}/\mathbb{Z}).$$

A map is monic if and only if its character dual is epic: $\mathbb{Q}/\mathbb{Z}$ is injective and detects nonzero abelian groups. More generally a sequence is exact if and only if its dual sequence is exact with arrows reversed.

Proposition. The following conditions on N are equivalent:

1. N is flat.
2. N^* is injective as a right module.
3. $I \otimes_A N \rightarrow N$ is injective for every right ideal $I \subseteq A$; its image is IN .
4. $\text{Tor}_1^A(A/I, N) = 0$ for every right ideal I .

For (1) and (2), use the adjunction

$$\text{Hom}_{A^{\text{op}}}(M, N^*) \simeq (M \otimes_A N)^*.$$

For a monomorphism $M \hookrightarrow M'$, extension of maps into N^* is equivalent, by character duality, to injectivity after tensoring:

$$\begin{array}{ccc} \text{Hom}_{A^{\text{op}}}(M', N^*) & \longrightarrow & \text{Hom}_{A^{\text{op}}}(M, N^*) \\ \sim \downarrow & & \downarrow \sim \\ (M' \otimes_A N)^* & \longrightarrow & (M \otimes_A N)^* \end{array}$$

Baer's criterion reduces this test to $I \hookrightarrow A$, giving (3). The Tor sequence for $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$ gives the equivalence with (4).

3.32 Finite presentation and projectivity

Definition. A module M is finitely presented if it admits an exact sequence $A^m \rightarrow A^n \rightarrow M \rightarrow 0$ with m, n finite. For an epimorphism $F \rightarrow M$ from a finitely generated module onto a finitely presented one, the kernel is finitely generated: lift a finite free presentation and compare its relations with a finite generating set of F .

For left modules M, N , evaluation gives a natural map

$$N^* \otimes_A M \longrightarrow \operatorname{Hom}_A(M, N)^*, \quad f \otimes m \mapsto (h \mapsto f(h(m))).$$

It is an isomorphism when M is finitely presented. First check $M = A^n$. Then take a finite presentation of M ; the two constructions give right exact rows, and the isomorphisms on the finite free terms identify their cokernels.

$$\begin{array}{ccccccc} N^* \otimes_A A^m & \longrightarrow & N^* \otimes_A A^n & \longrightarrow & N^* \otimes_A M & \longrightarrow & 0 \\ \sim \downarrow & & \downarrow \sim & & \downarrow & & \\ \operatorname{Hom}_A(A^m, N)^* & \longrightarrow & \operatorname{Hom}_A(A^n, N)^* & \longrightarrow & \operatorname{Hom}_A(M, N)^* & \longrightarrow & 0 \end{array}$$

Theorem. *A finitely presented flat module is projective.*

Proof. For an epimorphism $N_1 \twoheadrightarrow N_2$, character duality gives a monomorphism $N_2^* \hookrightarrow N_1^*$. Tensoring with the flat M preserves it. The preceding isomorphism identifies this with

$$\operatorname{Hom}_A(M, N_2)^* \hookrightarrow \operatorname{Hom}_A(M, N_1)^*.$$

Dual detection of exactness implies $\operatorname{Hom}_A(M, N_1) \twoheadrightarrow \operatorname{Hom}_A(M, N_2)$ is surjective. Thus M has the projective lifting property.

$$\begin{array}{ccccccc} 0 & \longrightarrow & N_2^* \otimes_A M & \longrightarrow & N_1^* \otimes_A M & & \\ & & \sim \downarrow & & \downarrow \sim & & \\ 0 & \longrightarrow & \operatorname{Hom}_A(M, N_2)^* & \longrightarrow & \operatorname{Hom}_A(M, N_1)^* & & \end{array}$$

□

3.33 Tor and flat base change

Proposition. A flat resolution computes Tor, because flat modules are acyclic for tensor. If $F \rightarrow M$ or $F' \rightarrow N$ is a flat resolution, then

$$\mathrm{Tor}_n^A(M, N) \simeq H_n(F \otimes_A N) \simeq H_n(M \otimes_A F').$$

For a ring map $A \rightarrow B$ with B flat as a left A -module, a right A -module M , and a left B -module N ,

$$\mathrm{Tor}_n^A(M, N) \simeq \mathrm{Tor}_n^B(M \otimes_A B, N).$$

Indeed a projective resolution $P \rightarrow M$ becomes the projective resolution $P \otimes_A B \rightarrow M \otimes_A B$, and the tensor complexes agree by associativity.

For a flat map of commutative rings $A \rightarrow B$, this implies

$$B \otimes_A \mathrm{Tor}_n^A(M, N) \simeq \mathrm{Tor}_n^B(B \otimes_A M, B \otimes_A N).$$

Multiplication by a central a on a module lifts to multiplication by a on a projective resolution, so induces multiplication by a on Tor. Thus if $aM = 0$, then $a \mathrm{Tor}_n^A(M, N) = 0$.

$$\begin{array}{ccc} P_{\bullet} & \xrightarrow{\times a} & P_{\bullet} \\ \downarrow & & \downarrow \\ M & \xrightarrow{\times a} & M \end{array}$$

For commutative A , localizing gives

$$\mathrm{Tor}_n^A(M, N)_{\mathfrak{p}} \simeq \mathrm{Tor}_n^{A_{\mathfrak{p}}}(M_{\mathfrak{p}}, N_{\mathfrak{p}}).$$

Consequently its vanishing can be tested at all primes, or just all maximal ideals.

3.34 Ext over the integers and behavior under products

For abelian groups, $\text{Ext}_{\mathbb{Z}}^n(A, B) = 0$ for $n \geq 2$: embed B into a divisible group I^0 ; the quotient is divisible and therefore injective, giving an injective resolution of length one.

For $r > 0$, the two-term free resolution of $\mathbb{Z}/r$ gives

$$\text{Ext}_{\mathbb{Z}}^n(\mathbb{Z}/r, B) = \begin{cases} B[r] & n = 0, \\ B/rB & n = 1, \\ 0 & n \geq 2. \end{cases}$$

For torsion A , use $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ and $\text{Hom}(A, \mathbb{Q}) = 0$ to obtain

$$\text{Ext}_{\mathbb{Z}}^1(A, \mathbb{Z}) \simeq A^* = \text{Hom}(A, \mathbb{Q}/\mathbb{Z}),$$

with all other degrees zero.

Proposition. For any ring and every n ,

$$\text{Ext}_A^n\left(\bigoplus_{\alpha} M_{\alpha}, N\right) \simeq \prod_{\alpha} \text{Ext}_A^n(M_{\alpha}, N),$$

$$\text{Ext}_A^n\left(M, \prod_{\beta} N_{\beta}\right) \simeq \prod_{\beta} \text{Ext}_A^n(M, N_{\beta}).$$

Use sums of projective resolutions in the first variable or products of injective resolutions in the second; products of module sequences are exact, so cohomology commutes with these products.

3.35 Scalar actions and localization of Ext

Let A be commutative. Ext groups are A -modules. Multiplication by a in either variable induces multiplication by a on $\text{Ext}_A^n(M, N)$: lift the scalar map to a projective or injective resolution. In particular $aM = 0$ or $aN = 0$ implies $a \text{Ext}_A^n(M, N) = 0$.

For a multiplicative set S , localization gives a natural map

$$S^{-1} \text{Hom}_A(M, N) \longrightarrow \text{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N).$$

It is an isomorphism when M is finitely presented. The result is immediate for a finite free module; a finite presentation reduces the general case to a kernel comparison, using exactness of localization.

Proposition. If A is noetherian and M finitely generated, choose a resolution by finitely generated free modules. Every successive kernel is finitely generated, so such a resolution exists in all degrees. Localize it and apply the Hom isomorphism termwise. For the finite free resolution $P_\bullet \rightarrow M$, the termwise comparison is:

$$\begin{array}{ccc} S^{-1} \text{Hom}_A(P_i, N) & \longrightarrow & S^{-1} \text{Hom}_A(P_{i+1}, N) \\ \sim \downarrow & & \downarrow \sim \\ \text{Hom}_{S^{-1}A}(S^{-1}P_i, S^{-1}N) & \longrightarrow & \text{Hom}_{S^{-1}A}(S^{-1}P_{i+1}, S^{-1}N) \end{array}$$

Exactness of localization then gives

$$S^{-1} \text{Ext}_A^n(M, N) \simeq \text{Ext}_{S^{-1}A}^n(S^{-1}M, S^{-1}N)$$

for every N and every n .

3.36 Extensions classified by the first Ext group

Definition. An extension of M by N is a short exact sequence $\xi : 0 \rightarrow N \rightarrow X \rightarrow M \rightarrow 0$. Two extensions are equivalent if a map of short exact sequences is the identity on the endpoints; the middle map is then an isomorphism. The split extension has middle term $N \oplus M$.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N & \longrightarrow & X & \longrightarrow & M & \longrightarrow & 0 \\ & & \parallel & & \downarrow \sim & & \parallel & & \\ 0 & \longrightarrow & N & \longrightarrow & X' & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

The connecting map for $\text{Hom}_A(M, -)$ associates

$$\Theta(\xi) = \delta(\text{id}_M) \in \text{Ext}_A^1(M, N).$$

It depends only on the equivalence class and vanishes precisely when id_M lifts to X , that is, when the extension splits.

Theorem. Θ is a bijection from equivalence classes of extensions of M by N to $\text{Ext}_A^1(M, N)$.

Choose $0 \rightarrow K \xrightarrow{j} P \rightarrow M \rightarrow 0$ with P projective. Every Ext class is represented by a map $u : K \rightarrow N$, modulo maps extending to P . Its extension is the pushout

$$X = (N \oplus P) / \{(u(k), -j(k)) : k \in K\}.$$

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K & \xrightarrow{j} & P & \longrightarrow & M & \longrightarrow & 0 \\ & & u \downarrow & & \downarrow & & \parallel & & \\ 0 & \longrightarrow & N & \longrightarrow & X & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

Proof. Conversely, lift $P \rightarrow M$ into any extension and restrict to K . Different lifts change u by a map from P , and equivalent extensions give the same class. These constructions are inverse, proving injectivity as well as surjectivity.

For $M = N = \mathbb{Z}/p$ there are p classes: one split class and $p - 1$ nonsplit classes, each with middle group $\mathbb{Z}/p^2$ but different endpoint identifications. Thus an extension class remembers the maps, not only the middle group.

3.37 Baer sums and Yoneda Ext

For extensions X, X' of M by N , form the fiber product over M and quotient by the anti-diagonal copy of N :

$$X'' = (X \times_M X') / \{(i(n), -i'(n)) : n \in N\}.$$

Its induced extension is the Baer sum.

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N \oplus N & \longrightarrow & X \times_M X' & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow + & & \downarrow & & \parallel & & \\ 0 & \longrightarrow & N & \longrightarrow & X'' & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

Equivalence classes form an abelian group with the split extension as zero, and Θ identifies this operation with addition in $\text{Ext}_A^1(M, N)$. Indeed lifts from a projective presentation add their maps $K \rightarrow N$ in exactly this construction.

In any abelian category, Yoneda defines n -extensions as exact sequences

$$0 \rightarrow N \rightarrow X_n \rightarrow X_{n-1} \rightarrow \cdots \rightarrow X_1 \rightarrow M \rightarrow 0,$$

modulo the equivalence relation generated by commutative maps of sequences identical on N, M . Baer addition is obtained by direct sum, pullback along the diagonal $M \rightarrow M \oplus M$, and pushout along addition $N \oplus N \rightarrow N$.

When there are enough projectives, these Yoneda groups agree with derived-functor Ext . Lift a projective resolution of M into an n -extension. The resulting map from the n th syzygy to N defines an Ext class by dimension shifting. Changing lifts produces the same class. Conversely pushout of the truncated projective resolution constructs an extension from such a map, and these operations are inverse. This generalizes the preceding degree-one classification. Here $\Omega^n M$ is the n th syzygy:

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & \Omega^n M & \longrightarrow & P_{n-1} & \longrightarrow & \cdots & \longrightarrow & P_0 & \longrightarrow & M & \longrightarrow & 0 \\ & & \downarrow u & & \downarrow & & & & \downarrow & & \parallel & & \\ 0 & \longrightarrow & N & \longrightarrow & X_n & \longrightarrow & \cdots & \longrightarrow & X_1 & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

3.38 Derived inverse limits of towers

Proposition. If $\mathcal{A}$ is complete and has enough injectives, so does $\mathcal{A}^I$ for a small category I . Evaluation at k is exact and has right adjoint

$$(R_k Q)_i = \prod_{\text{Hom}_I(i, k)} Q.$$

Choose $X_k \hookrightarrow Q_k$ injective. Adjunction gives a monomorphism $X \hookrightarrow \prod_k R_k Q_k$, with injective target. The identity coordinate at each k detects injectivity.

The constant-diagram functor is left adjoint to inverse limit, so inverse limit is left exact and admits right derived functors. For a countable tower of abelian groups

$$\cdots \xrightarrow{f_3} A_2 \xrightarrow{f_2} A_1 \xrightarrow{f_1} A_0,$$

define

$$\Delta : \prod_{i \geq 0} A_i \rightarrow \prod_{i \geq 0} A_i, \quad \Delta(a)_i = a_i - f_{i+1}(a_{i+1}).$$

Then $\varprojlim A_i = \ker \Delta$, $\varprojlim^1 A_i = \text{coker } \Delta$, and $\varprojlim^n = 0$ for $n > 1$. The snake lemma on the two product rows of a short exact sequence of towers makes these a cohomological δ -functor.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \prod_i A_i & \longrightarrow & \prod_i B_i & \longrightarrow & \prod_i C_i \longrightarrow 0 \\ & & \Delta_A \downarrow & & \downarrow \Delta_B & & \downarrow \Delta_C \\ 0 & \longrightarrow & \prod_i A_i & \longrightarrow & \prod_i B_i & \longrightarrow & \prod_i C_i \longrightarrow 0 \end{array}$$

Proposition. If every f_i is onto, then Δ is onto: solve recursively for a_{i+1} from the desired i th component, starting with any a_0 . Such towers have vanishing $\varprojlim^1$.

3.39 Effacement and the Mittag-Leffler condition

Definition. An additive functor F is effaceable if every object X admits a monomorphism $X \hookrightarrow I$ on which the induced map under F is zero. A cohomological δ -functor is universal when its positive degrees are effaceable. To extend a map in degree zero, use $0 \rightarrow X \rightarrow I \rightarrow I/X \rightarrow 0$: the vanishing map makes the connecting morphism onto, and its universal factorization determines the next natural transformation. Induction gives all degrees and uniqueness. For a morphism $T^\bullet \rightarrow F^\bullet$ defined below degree n , choose $X \hookrightarrow I$ making $T^n(X) \rightarrow T^n(I)$ zero:

$$\begin{array}{ccccccc} T^{n-1}(I) & \longrightarrow & T^{n-1}(I/X) & \xrightarrow{\partial} & T^n(X) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ F^{n-1}(I) & \longrightarrow & F^{n-1}(I/X) & \xrightarrow{\partial} & F^n(X) & & \end{array}$$

Proposition. Every tower embeds in a tower with surjective transition maps, using the product of the right-adjoint diagrams $R_k Q_k$. Hence the preceding $\varprojlim^1$ is effaceable and computes $R^1 \varprojlim$; all higher derived limits of countable towers vanish.

Definition. A tower is Mittag-Leffler if, for each k , the images of $A_i \rightarrow A_k$ stabilize as $i \rightarrow \infty$. Write B_k for the stable image. The tower (B_k) has onto transitions, and the quotient tower has eventually zero maps into each fixed level. For that quotient, solve $\Delta(a) = b$ by the finite sum

$$a_k = b_k + f_{k+1}(b_{k+1}) + f_{k+1}f_{k+2}(b_{k+2}) + \cdots ;$$

only finitely many terms are nonzero for each k . Thus $\varprojlim^1 A_i = 0$ for any Mittag-Leffler tower.

The tower $p^i \mathbb{Z}$ with inclusion maps need not have this property. From $0 \rightarrow p^i \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}/p^i \mathbb{Z} \rightarrow 0$ one obtains

$$\varprojlim^1 p^i \mathbb{Z} \simeq \mathbb{Z}_p / \mathbb{Z}.$$

3.40 Milnor sequences and a topological application

For a degreewise Mittag-Leffler tower of chain complexes C_i , let $C = \varprojlim C_i$. There is a natural exact sequence

$$0 \rightarrow \varprojlim^1 H_{n+1}(C_i) \rightarrow H_n(C) \rightarrow \varprojlim H_n(C_i) \rightarrow 0.$$

One proof uses the degreewise surjectivity of Δ to form the short exact sequence of complexes

$$0 \rightarrow C \rightarrow \prod_i C_i \xrightarrow{\Delta} \prod_i C_i \rightarrow 0.$$

Products of abelian groups are exact, so their homology is the product of the homologies; the associated long exact sequence yields the displayed kernel-cokernel sequence.

For cochain complexes this reads

$$0 \rightarrow \varprojlim^1 H^{n-1}(C_i) \rightarrow H^n(\varprojlim C_i) \rightarrow \varprojlim H^n(C_i) \rightarrow 0.$$

Applied to an increasing union of CW complexes $X = \cup_i X_i$, the corresponding Milnor sequence is

$$0 \rightarrow \varprojlim^1 H^{n-1}(X_i; \mathbb{Z}) \rightarrow H^n(X; \mathbb{Z}) \rightarrow \varprojlim H^n(X_i; \mathbb{Z}) \rightarrow 0.$$

For $\mathbb{CP}^\infty = \cup_i \mathbb{CP}^i$, the rings at finite stages are $\mathbb{Z}[t]/(t^{i+1})$, with $|t| = 2$, and each fixed cohomological degree stabilizes. Thus $H^*(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}[t]$ as a graded ring.

3.41 Universal coefficients for homology

Let P be a chain complex of right A -modules such that all P_n and $B_n(P)$ are flat. Then $Z_n(P)$ is flat as well, by $0 \rightarrow Z_n \rightarrow P_n \rightarrow B_{n-1} \rightarrow 0$. For every left module M there is a natural exact sequence

$$0 \rightarrow H_n(P) \otimes_A M \rightarrow H_n(P \otimes_A M) \rightarrow \operatorname{Tor}_1^A(H_{n-1}(P), M) \rightarrow 0.$$

To prove it, tensor the short exact sequence of complexes $0 \rightarrow Z(P) \rightarrow P \rightarrow B(P)[-1] \rightarrow 0$. Flatness of the boundaries preserves exactness. Its homology sequence identifies the left term with the cokernel of $B_n \otimes M \rightarrow Z_n \otimes M$ and the right term with the kernel of $B_{n-1} \otimes M \rightarrow Z_{n-1} \otimes M$. The flat resolution $0 \rightarrow B_{n-1} \rightarrow Z_{n-1} \rightarrow H_{n-1} \rightarrow 0$ identifies that kernel with Tor .

Proposition. If $A = \mathbb{Z}$ and P consists of free abelian groups, the sequence splits, generally noncanonically:

$$H_n(P \otimes M) \simeq H_n(P) \otimes M \oplus \operatorname{Tor}_1^{\mathbb{Z}}(H_{n-1}(P), M).$$

For the singular chain complex $S(X)$, $S_n(X)$ is free on continuous maps $\Delta^n \rightarrow X$, with boundary the alternating sum of face restrictions. Thus the formula computes homology with coefficients M from integral homology.

3.42 Kunneth and universal coefficients for cohomology

For chain complexes P of right and Q of left modules, assume P_n and its boundaries are flat. These hypotheses are essential over a general ring; without them the result is normally expressed by a Kunneth spectral sequence. The tensor differential is

$$d(a \otimes b) = da \otimes b + (-1)^{\deg a} a \otimes db.$$

The same argument with $0 \rightarrow Z(P) \otimes Q \rightarrow P \otimes Q \rightarrow B(P)[-1] \otimes Q \rightarrow 0$ gives the Kunneth sequence

$$\begin{aligned} 0 \rightarrow \bigoplus_{p+q=n} H_p(P) \otimes_A H_q(Q) &\rightarrow H_n(P \otimes_A Q) \\ &\rightarrow \bigoplus_{p+q=n-1} \operatorname{Tor}_1^A(H_p(P), H_q(Q)) \rightarrow 0. \end{aligned}$$

Over $\mathbb{Z}$, if P is free, it splits noncanonically. The Eilenberg–Zilber equivalence $S(X \times Y) \simeq S(X) \otimes S(Y)$ then gives the usual Kunneth formula for the homology of a product of spaces.

Proposition. If P is a complex of projective left A -modules whose boundaries are projective, the cohomological universal coefficient sequence is

$$\begin{aligned} 0 \rightarrow \operatorname{Ext}_A^1(H_{n-1}(P), M) &\rightarrow H^n \operatorname{Hom}_A(P, M) \\ &\rightarrow \operatorname{Hom}_A(H_n(P), M) \rightarrow 0. \end{aligned}$$

It splits noncanonically. Apply Hom to the degreewise split sequence $0 \rightarrow Z(P) \rightarrow P \rightarrow B(P)[-1] \rightarrow 0$, then identify the kernels and cokernels using the projective presentations $0 \rightarrow B_n \rightarrow Z_n \rightarrow H_n \rightarrow 0$.